

Getting Started V3 — Minecraft 1.21.1 NeoForge

This guide walks you through the modpack **one Wave at a time**. Each wave introduces one new capability. Don't install the next wave until you're comfortable with the current one.

Wave -1 — How to Play Minecraft

This section assumes you know nothing. If you've played Minecraft before, skip to Wave 0.

The Goal

There isn't one. Minecraft is a sandbox. You punch trees, build things, explore, fight monsters, and eventually become very powerful. You make your own goals. This modpack adds even more things to do on top of the base game.

Controls (Default)

Action	Key
Move	W A S D
Jump	Space
Sneak	Shift (hold to walk off edges without falling)
Sprint	Ctrl (hold while moving)
Attack / Break blocks	Left-click (hold to keep breaking)
Use item / Place block	Right-click
Open inventory	E
Drop item	Q (while hovering over item in inventory)
Select hotbar slot	1 through 9
Sneak + right-click on chest	Opens chest without placing the block you're holding

Customize these: Options → Controls → Key Binds. Set what feels natural. The most common change is swapping sprint to a toggle.

Your First 10 Minutes

1. Punch a tree

Look at a tree trunk. **Hold left-click** until the wood block breaks and drops. Pick it up by walking over it. Get at least 8 wood logs.

2. Make a crafting table

Open your inventory (E). You'll see a 2x2 crafting grid in the top-right. Place 1 wood log in any slot → 4 wood planks appear on the right. Take them. Then fill all 4 crafting slots with planks (put 1 plank in each) → 1 crafting table appears. Take it.

Quick tip: In your inventory grid, you can also turn 2 planks into 4 sticks (plank on top, plank on bottom). Sticks are used for tool handles.

3. Place your crafting table

Walk to where you want your base. Right-click the ground with the crafting table in your hand → it appears. **Right-click the table** to open its 3x3 crafting grid (the big one).

Now craft tools:

Tool	What it does	Crafting (3x3 grid)
Wooden Pickaxe	Breaks stone	3 planks across top row, 2 sticks down the middle column
Wooden Axe	Faster tree chopping	3 planks in top-left corner, 2 sticks down
Wooden Shovel	Digs dirt/sand faster	1 plank top-center, 2 sticks below
Wooden Sword	Fights mobs	2 planks in middle column, 1 stick below

4. Gather stone

Find some grey stone on the ground (or cliffs). Use your **wooden pickaxe** to mine 8-12 stone blocks. Then craft **stone tools** — same patterns as wooden but replace planks with cobblestone. Stone tools last longer and mine faster.

5. Build a shelter

Before night falls (about 10 minutes into the first day), build a small box:

- Walls: dirt, wood planks, or cobblestone — whatever you have
- Roof: same material
- Door: craft from 6 planks (2 columns of 3 in the crafting grid)
- Torches: 1 stick + 1 coal (or charcoal) = 4 torches. Place them inside so you can see.

No coal? You can make **charcoal** by smelting wood logs in a furnace (see below). Same as coal but from wood.

6. Craft a furnace

Material	Pattern
8 cobblestone	All slots except the center one → furnace

Place the furnace. Right-click it. Put wood logs in the top slot, planks/sticks in the bottom slot (fuel). Smelt logs into charcoal, cook raw meat, smelt iron ore.

7. Survive the first night

Monsters spawn in darkness. Your shelter keeps them out. Stay inside. You can:

- Dig a small mine under your shelter
- Expand your crafting table base
- Smelt materials
- Sleep if you have a bed (3 wool + 3 planks)

If there's no bed: Wool comes from killing sheep (don't kill all the sheep — breed them with wheat later). Different coloured sheep drop different wool. 3 of any colour + 3 planks in a row = bed.

8. Day 2 and beyond

Priority	What to do
Find food	Kill animals (cows, pigs, chickens). Cook the meat in your furnace. Plant wheat seeds (break grass) near water.
Get iron	Dig down (not straight down — staircase pattern) until you see brownish blocks with specks. Mine with stone pickaxe. Smelt in furnace → iron ingots.
Make iron tools	Iron pickaxe, iron sword, iron armour. This is the "I'm established" checkpoint.
Find a village	Walk around until you see buildings with paths. Villages have beds, food crops, and later — Waystones.
Make a bed	3 wool + 3 planks. Sleep through the night. Respawn at bed if you die.

Navigating with F3

Press **F3** opens the debug screen. The useful parts:

Value	What it shows
xyz	Your coordinates — write these down so you can find your base
Biome	What biome you're in
Direction	Which way you're facing (N/S/E/W)
Light	Light level — mobs spawn at light level 0

Quick tip: Write down your base coordinates before exploring. Or just place a Waystone (Wave 0) at your base.

Dying

You lose some or all of your items (depends on difficulty settings) and respawn at your bed (or world spawn if no bed). With Wave 0's **grave mod**, your items stay in a grave you can retrieve.

Key Minecraft Concepts

Health & food: You have 10 hearts (20 HP). Food restores hunger. When hunger is full, you slowly regenerate health. Don't let hunger drop to 0 — you'll start taking damage.

Day/night cycle: 20 minutes total. 10 minutes day, 1.5 minutes sunset, 7 minutes night, 1.5 minutes sunrise. Monsters spawn in darkness.

Ore tiers: Wood → Stone → Iron → Diamond → Netherite. Each tier mines faster and can mine the next tier's ores.

The Nether: A dangerous alternate dimension. Build a Nether Portal (10 obsidian blocks in a 4x5 rectangle, light with flint and steel). The Nether has unique resources (blaze rods, netherite) and is optional for most of this modpack.

The End: The dimension with the Ender Dragon. Beating the dragon is the "credits roll" of vanilla Minecraft. This modpack has many more things to do after that.

Wave 0 — Foundation

This wave makes Minecraft run well, look great, and feel comfortable. No major gameplay changes — just setting up your world to be livable.

Install

See [MODLIST-V3.md](#) for the full mod list with download links. This is a **NeoForge** loader pack — install NeoForge, not Fabric.

First Launch Settings

Video Settings

Setting	Recommended	Notes
Graphics	Fabulous	Sodium makes this performant now
Render Distance	16-32 chunks	Tune to your hardware; Distant Horizons handles far LOD
Brightness	Moody → Bright	Personal preference — Bright makes caves easier to see

Distant Horizons Settings

Open **Mod Menu** → **Distant Horizons** → **Config** after first launch. Set LOD Render Distance to 1024 blocks. The mod generates LOD data as you explore — first visit to an area will have temporary pop-in.

Performance — Install and Forget

These mods work automatically with no configuration. Install them and they optimize rendering, chunk generation, and entity processing in the background:

Mod	What it does
Radium	Server-side optimization — faster entity AI, block updates, and chunk physics
ModernFix	Fixes performance bugs and memory leaks from other mods. All-in-one optimization
ImmediatelyFast	Speeds up block and item rendering by using immediate-mode rendering where possible
Entity Culling	Stops rendering entities that are behind walls or off-screen. Free FPS
Dynamic FPS	Reduces frame rate when Minecraft is in the background — saves GPU power
NoisiumForked	Accelerates world generation — faster chunk creation when exploring

No config needed for any of these. Install, launch, forget.

Shader Setup

Place **Complementary Unbound** .zip in the `shaderpacks/` folder. Launch Minecraft, go to Options → Video Settings → Shaderpacks, select Complementary Unbound. It works with Iris on NeoForge.

Texture Packs

Two optional resource packs enhance vanilla visuals without changing the game feel:

Pack	Role
Faithful 64x	64x upscaled vanilla textures — clean, sharp, PBR-compatible with Complementary Unbound
Fresh Animations	Living entity animation overhaul — mobs blink, look around, show emotion

Installation: Download the .zip files, place them in `.minecraft/resourcepacks/`, enable both in Options → Resource Packs (Fresh Animations above Faithful 64x). Faithful 64x works with Complementary Unbound's Integrated PBR+ for realistic material response. Modded blocks (Create, Mekanism, AE2) use their default textures which blend naturally at 64x.

Keybindings (Set These Up First)

Keybind	Default	Mod	Notes
View Recipes	R	JEI	Hover over any item and press R
View Uses	U	JEI	Hover over any item and press U
Open Fullscreen Map	J	Xaero's World Map	Opens the big map
Toggle Minimap	H (default)	Xaero's Minimap	Shows/hides the corner minimap
Open Waystone Menu	Not set	Waystones	Set to Ctrl+Shift+W
Open Backpack	B	Sophisticated Backpacks	Opens equipped backpack

Keybind	Default	Mod	Notes
Zoom	Not set	Sodium	Set to C — hold to zoom in

Session Plan — Wave 0

Your first session is about getting comfortable. Don't rush — this wave has zero progression pressure.

Session	Goal
Session 1 — Setup	Launch the game. Set keybinds (Ctrl+Shift+W for Waystones, C for zoom). Open Mod Menu to confirm all mods loaded. Configure Distant Horizons LOD to 1024. Walk around spawn to generate LOD data
Session 2 — Survival basics	Punch trees, build a starter shack, find food, get iron. Place your first Waystone at your base. Craft a Sophisticated Storage barrel for organized storage. Craft a backpack, equip it, set hotkey
Session 3 — Comfort	Explore nearby terrain. Find a village. Set up a bed. Install the Complementary Unbound shaderpack if desired. Make torches and light up your base perimeter
Ongoing	Harvest With Ease works as you farm. LambDynamicLights lights your way as you hold a torch. AmbientSounds plays nature music automatically. Chunk Loaders wait until you have automation to protect — place one at your base and forget it

Using Wave 0 Mods

JEI (Recipe Viewer)

Your primary learning tool. Hover any item and press R to see how to craft it, U to see what it's used in. Every mod in future waves is learned through JEI first.

Jade + Jade Addons (Block Info)

Jade shows what block you're looking at and extra information about it — modded machine status, crop growth stage, chest contents, and more.

How to use:

- Look at any block. A HUD panel shows the block name, mod source, and relevant data
- Press the keybind (Controls → Jade → Open Config, or hold the key while looking at a block) to adjust what information is shown
- **Jade Addons** extends Jade with panels for Create (SU/RPM), Mekanism (energy/gas), AE2 (channel info), and other modded machines

Pro tip: Keep Jade enabled. It's the fastest way to check if a machine is running, a crop is ready, or a chest contains specific items.

AppleSkin (Food Info)

AppleSkin adds a visual overlay to your hunger bar showing exactly how much hunger and saturation food will restore before you eat it.

What it shows:

- **Hover over food** in your inventory or JEI — see exact hunger and saturation values
- **Pre-eat preview** — your hunger bar shows dashed outlines for what the food will fill
- **Saturation overlay** — a second layer on the hunger bar shows your hidden saturation stat

No config needed — install and forget.

Polymorph (Recipe Conflicts)

When two mods have different recipes for the same item (common in a 112-mod pack), Polymorph lets you choose which recipe to use at the crafting table.

How it works:

1. Place ingredients in the crafting grid as usual
2. If multiple recipes match, a small selection button appears next to the output slot
3. Click it to cycle through available recipes and pick the result you want
4. The selection persists for that ingredient combination until you change it

No config needed — works automatically. Most relevant when crafting basic materials like planks, ingots, or components that multiple mods define.

Sophisticated Storage & Backpacks

Craft a barrel (1 chest + 8 planks) for early storage. Upgrade with iron/gold/diamond/netherite upgrades. Backpack in your chest armor slot, press **B** to open.

Waystones

Craft a waystone and right-click to claim it. Use the Warp Stone (ender pearl + amethyst) to teleport from anywhere. No XP costs. Set keybind to **Ctrl+Shift+W**.

GraveStone Mod

When you die, a grave spawns with your items. No item loss, no despawning. Just walk back and right-click the grave.

AmbientSounds + The Immersive Music Mod

AmbientSounds adds nature sounds (birds, wind, water) based on your biome and time of day. [The Immersive Music Mod](#) adds 147 new songs selected by biome and structure — plays like vanilla with natural silence gaps, no config needed.

BetterF3

Press **F3** to see the cleaner debug HUD. Customize what's shown in Mod Menu → BetterF3 → Config. Toggle individual modules (FPS, coordinates, biome, light level) without the clutter.

LambDynamicLights

Equip a torch or glowstone block → it emits light around you. Works in the off-hand too. Toggle via keybind **Ctrl+L**.

Pro tip: Hold a torch in your off-hand while mining — no more placing torches every 5 blocks.

Sound Physics Remastered

Makes sound behave more realistically — muffled through walls, echo in caves, directional audio. Works automatically with any sound pack.

Mouse Tweaks

Inventory management shortcuts: left-click drag across slots → distributes evenly. Right-click drag → places one per slot. Scroll wheel on a stack → moves one to another inventory.

Curios API (Accessory Slots)

Opens extra equipment slots on your character: ring, amulet, belt, gloves, charm, back, and more. Press the Curios keybind (check Controls) to open the Curios inventory. This replaces Trinkets from V2 — Relics RPG accessories and other Curios-compatible items go in these slots.

Reliable Advancements

Reliable Advancements overhauls the vanilla advancements screen — larger UI, pan/zoom controls, tab sorting, and a built-in advancement editor. Open it from the usual advancements key (**L** by default). Pairs well with Fancy Toasts (the animated popups you see when earning an advancement).

No configuration needed. Install and forget.

BlazeandCave's Advancements Pack (Data Pack)

This data pack adds **1000+ new advancements** across 16 tabs — mining, building, farming, combat, exploration, redstone, brewing, enchantments, and more. It tracks milestones like mining every ore type, collecting every wood variant, killing every mob, and discovering every biome.

Where to install: Drop the **.zip** into the **datapacks/** folder of your world save (**.minecraft/saves/<world>/datapacks/**). Enable it with **/datapack enable "blazeandcave/advancements"**. It works alongside Reliable Advancements (UI) and Fancy Toasts (popups) without conflict.

What it adds: Thousands of hours of optional side goals. Everything is tracked through the vanilla advancement system — open with **L** to browse the 16 tabs and see what's available.

Loot Journal: Pickup Notifier

Shows animated pickup notifications when items enter your inventory and tracks session statistics.

What it does:

- **Pickup popups** — items you pick up appear as animated icons near your hotbar with quantity counters
- **Session summary** — when you disconnect, a summary shows what you gained and lost during the session
- **Statistics** — cumulative drop tracking across all sessions

Config (via Mod Menu → Loot Journal):

- `writeSessionJson = true` — Saves per-session gain/loss data (default)
- `writeStatisticsJson = true` — Builds cumulative drop statistics (default)
- `showSummaryOnDisconnect = true` — Shows session summary on disconnect (default)

Default settings work well — no changes needed.

Harvest With Ease

Right-click a fully grown crop → it harvests and auto-replants. Works on vanilla crops and most modded ones.

Immersive UI

Immersive UI adds subtle animation and polish to the game interface — the hotbar slides smoothly, items have gentle particle effects, and menus feel responsive rather than instant-teleport.

What it changes:

- Smooth hotbar selection sliding
- Item pickup particles in the UI
- Animated tooltips and menu transitions
- Crafting grid visual feedback

No config needed — install and forget.

Traveler's Titles

Displays an RPG-style title card when you enter a new biome or dimension — large text with a subtitle fades in, lingers, and fades out.

What it shows:

- **Biome entry** — name of the biome with a thematic subtitle
- **Dimension entry** — dimension name with an atmospheric description (e.g., "The Nether — A Hellish Landscape")
- **Height change** — entering deep caves or high mountains triggers a title

No config needed. If titles feel too frequent, adjust display time in Mod Menu → Traveler's Titles → Config.

Better Modlist

Overhauls the Mod Menu list screen with badges, categories, and search — makes finding and configuring mods much faster.

What it adds:

- **Badges** — visual tags showing mod type (library, core, addon, utility)
- **Categories** — mods grouped by role
- **Search** — type to filter the mod list
- **Improved layout** — more mods visible per page, cleaner scrolling

No config needed — install and forget.

Better Third Person

Better Third Person decouples your camera from your character in third-person view. You can look around independently while your character walks in any direction — essential for combat awareness and scenic building.

How to use:

- Press the toggle keybind (default: not set — bind it in Controls → Better Third Person)
- Once active, your character model moves independently of the camera
- Great for combat: look behind you while running away
- Great for building: orbit the camera around your build while placing blocks
- Works with Eating Animation — you see your character eat in 3rd person

No config changes needed — default settings work well.

Tooltip Overhaul

Tooltip Overhaul redesigns the tooltip appearance — rounded corners, smoother background, and better text contrast. It works alongside Obscure Tooltips (animated visual effects) without conflict — Tooltip Overhaul handles the background/layout layer, Obscure Tooltips adds the animated frame/glow layer.

No config needed — install and forget.

Eating Animation

Adds a visible eating animation in both first-person and third-person view. Food visibly shrinks as you eat. Works with Farmer's Delight foods out of the box — no extra config needed.

No config needed.

Mod Name Tooltip

Hover over any item in your inventory or JEI — the tooltip shows which mod added the item in small text. Helps you learn what belongs to which mod as you explore.

No config needed.

Chunk Loaders — Infrastructure

Chunk Loaders keep specific chunks loaded even when you're far away or in another dimension. Essential for keeping factories, farms, and your colony running.

Types of Loaders:

- **Single Chunk Loader** — keeps 1 chunk loaded forever. No fuel needed. Use to keep one machine or farm running
- **Spotloader** — handheld, temporarily loads an area while you explore nearby
- **Multi Chunk Loader** — loads a 3×3 chunk area. Use for your entire base or factory complex

How to use:

1. **Craft a Single Chunk Loader** (check JEI — requires iron, redstone, and ender pearl)
2. **Place it** at the center of the area you want to keep loaded
3. **Right-click** to toggle it on — a colored border shows the loaded chunk
4. **Done.** The chunk stays loaded across all dimensions

Best practices:

- Place a loader at your **main base** first — keeps all Create/Mek/AE2 systems running
- Place a loader at your **colony** (Wave 1.5) — keeps MineColonies workers active
- Place a loader at each **mining outpost** — keeps the Mekanism digital miner running
- Loaders work in any dimension. The Nether, The End, other planets — a loader keeps that chunk ticking

Tip: You can have as many loaders as you want. No performance penalty as long as your PC handles the loaded chunks. Start with 2-3 and add as needed.

Config: No changes needed. Default range (1 chunk per loader) and recipe costs are appropriate.

Wave 1 — Tech (Create + Mekanism + AE2)

The biggest wave. Three industry-standard mods that together form a complete tech ecosystem. Wave 1.5 (Colony Management) runs alongside this — start your colony after you have basic Create power.

Install

See [MODLIST-V3.md](#). Install all three core mods at once.

How They Fit Together

Problem	Solution
Early power	Create: water wheels, windmills
Ore processing	Mekanism: from 2x to 5x ore multiplication
Item transport	Create: conveyor belts, chutes, funnels
Factory automation	Create: deployers, sequenced crafters, mechanical arms
Digital storage	AE2: ME drives store thousands of items in one block

Problem	Solution
Autocrafting	AE2: pattern-provider autocrafting
Mid-game power	Mekanism: gas-burning generators, solar, wind
Endgame power	Mekanism: fusion reactor
Endgame armor	Mekanism: MekaSuit (flight, invulnerability, fire resistance)
Long-distance transport	Create: trains

Create — Mechanical Automation

Create is about **rotational power**. Every Create machine needs rotational force to work. You generate this force, transfer it through shafts and belts, and use it to power machines.

The Three Core Concepts

Concept	What it means	Why it matters
Rotational Power (SU)	Every Create source generates Stress Units (SU). Machines consume SU to run	If total SU consumption exceeds generation, machines stop. Add more sources or reduce load
Stress Capacity	Shafts and gearboxes can only handle so much SU before breaking	Use reinforced shafts for high-stress networks. Spread load across multiple lines
Speed vs Force	Speed (RPM) determines how fast a machine processes. Force determines how much load it can handle	Gear ratios matter: small gear on large gear = slow + strong. Large on small = fast + weak

Learning with Ponder

Create has a built-in interactive tutorial called the **Ponder system**. Hold **W** while hovering over any Create block or item in JEI or your inventory. An animated scene plays showing exactly how the block works, what it connects to, and what it does.

Use Ponder constantly. It is better than any written guide. Every Create player learns through Ponder first.

First Create Machines

Your goal: build a simple processing line for early materials.

Step 1 — Gather basic components:

- Mine **andesite** — the most important early resource. You need lots of it
- Craft **andesite casing** (andesite + wood nugget in JEI) — the basic building block of all Create machines
- Craft **cogwheel** and **large cogwheel** — transfer rotational power

Step 2 — Build a hand-cranked system:

1. Craft a **hand crank** — place it on the ground
2. Attach a **cogwheel** next to it — right-click the crank to spin it manually
3. Attach a **millstone** to the cogwheel — right-click the millstone, put wheat or ore in

This is your first Create setup. It's slow but it works without any power source.

Step 3 — Build a water wheel:

1. Place **water source blocks** flowing downhill (2-3 blocks drop)
2. Place a **water wheel** in the flowing water — it starts spinning
3. Attach a **shaft** to the wheel — rotational power flows through it
4. Use **gear boxes** to change direction or split power

A water wheel generates 256 SU when placed in flowing water. This is enough to power 2-3 basic machines.

Step 4 — Automate processing:

1. **Mechanical press** — press ingots into plates, press plates into mechanisms
2. **Millstone** — grind ores into dust (doubles output for smelting)
3. **Mechanical mixer** — mix ingredients into alloys and compounds

Connect each machine to your power line with cogwheels. Right-click with a **wrench** to adjust direction.

Mid-Game Create

Once you have brass (from zinc + copper in a heated mixer), you unlock:

Machine	What it does
Deployer	Places blocks or uses items from its inventory. Automates crafting
Sequenced crafter	Runs a multi-step crafting sequence automatically. Your first autocrafter
Mechanical arm	Moves items between inventories. Replaces conveyor belts
Chute & Funnel	Vertical item transport with filtering
Train station	Build tracks, assemble trains, transport bulk items between bases

Brass progression: Find zinc ore (generates in the world alongside copper) → Mix zinc + copper in a heated mixer → Brass ingots → Brass casing → Advanced machines.

Blueprint system: Craft a **blueprint and quill**. Select an area, save it as a blueprint. Place a **deployer** with the blueprint to automatically build it. This is how you replicate factory sections.

Create Keybindings

Keybind	Default	What it does
Open Ponder	W (hold over block)	Shows interactive animation for Create blocks
Open Toolbox	T (hold)	Quick-access tool menu

Keybind	Default	What it does
Wrench functionality	Right-click	Configure machine direction, dismantle
Goggles functionality	G (hold over machine)	Shows SU consumption, RPM, and stress info
Open Blueprint	Not set	Opens the blueprint editor

Goggles are essential. Craft a pair of engineers goggles early (check JEI). Hold G while looking at any Create machine to see its SU consumption, current RPM, and whether it's overstressed.

Create: The Factory Must Grow (TFMG) — Industrial Oil & Steel

TFMG extends Create into the oil age. Drill for crude oil, refine it into fuels, and burn them in diesel engines for massive rotational power. Produce steel at industrial scale and generate electricity with voltage management.

Core Concept — Oil Processing

Oil is the central resource of TFMG. You find it in the world, extract it, refine it, and use the byproducts.

Finding oil:

- Oil generates as underground pools (similar to Mekanism's oil, but separate)
- Find surface seepage in desert and plains biomes — dark patches on the ground
- Dig down to find oil-bearing sand/stone
- Use a **Pumpjack** (crafted from Create brass and steel) to extract oil automatically

Refining crude oil:

Step	Machine	Input	Output
1	Distillation Tower	Crude oil	Diesel, gasoline, LPG, lubricant, asphalt
2	Cracker	Heavy fractions	Lighter fuels
3	Desulfurizer	Diesel/gasoline	Clean fuel (more efficient)

Build a **Distillation Tower** (multi-block structure — minimum 4 blocks tall). Feed crude oil into the bottom. Products come out at different heights — lighter products (LPG, gasoline) exit higher, heavier products (diesel, lubricant) exit lower.

Diesel Engines — Massive Rotational Power

Diesel engines burn TFMG fuels to generate rotational power. A single diesel engine running on diesel fuel produces significantly more SU than any Create water wheel or windmill — enough to power an entire factory complex.

Engine	Fuel	SU Output	Best for
Small Diesel Engine	Diesel, gasoline	~2,000 SU	Early oil — single machine power

Engine	Fuel	SU Output	Best for
Large Diesel Engine	Diesel, gasoline	~8,000 SU	Factory-scale power
Gas Turbine	LPG	~16,000 SU	Endgame power generation

Fuel comparison:

Fuel	Efficiency	Engine type	Byproduct use
Diesel	High	Diesel engine	Most efficient general fuel
Gasoline	Medium	Diesel engine	Good backup fuel, less efficient
LPG	Very High	Gas turbine	Best power density, limited quantity
Lubricant	—	Not a fuel	Used in steelmaking and machines
Asphalt	—	Not a fuel	Decorative building blocks

Note: TFMG diesel engines are separate from Create's own steam engines. They provide more power per block but require the oil processing chain to run. Plan your progression: water wheel → steam engine → diesel engine.

Steel Blast Furnace

TFMG adds a multi-block **Steel Blast Furnace** for industrial-scale steel production. Unlike Mekanism's Metallurgic Infuser (which produces steel slowly), the Blast Furnace produces steel in bulk.

Setup:

1. Build the multi-block structure (check JEI/Ponder for the exact pattern)
2. Feed iron ingots and coal/coke into the top
3. Power with a diesel engine or Create rotational power
4. Liquid steel pours out the bottom — cast it into steel ingots or blocks

Why use it: Once you're producing oil and running diesel engines, the Blast Furnace produces steel faster than the Metallurgic Infuser. Use it to mass-produce steel for Mekanism machines, Create trains, and Stellaris rocket components.

Aluminum

TFMG adds aluminum as a new metal. **Bauxite ore** generates in the Overworld (typically at mid-to-high Y-levels). Process it through a multi-block smelter to produce aluminum ingots.

Uses:

- Aluminum is used in advanced TFMG machines
- Feeds into Mekanism's advanced alloy processing
- Used in some Stellaris rocket components (via TFMG-Stellaris Compatibility)

Progression Notes

- TFMG is a **Wave 1 addition** — start exploring oil after you have basic Create (water wheel + hand crank) and basic Mekanism (Enrichment Chamber)
- The Distillation Tower requires brass-tier Create components (brass casing, brass funnels)
- Diesel engines are a mid-to-late game power upgrade — they replace/supplement water wheels and windmills
- Mekanism TFMG Compatibility (installed alongside TFMG) resolves overlapping recipes between TFMG and Mekanism — see the config section below for details

Mekanism — Industrial Processing

Mekanism is about **processing materials at industrial scale**. Its power system (Joules) is separate from Create's rotational power, but you can bridge them — **Create Crafts & Additions** provides electric motors and alternators (convert between kinetic SU and Forge Energy FE), and Create's built-in alternator also works.

Finding Osmium

Osmium is Mekanism's primary ore. It generates underground at all levels, similar rarity to iron. Mine it with a stone pickaxe or better. Smelt raw osmium into osmium ingots.

The Tier System

Every Mekanism machine comes in 4 tiers:

Tier	Color	Processing Speed	Upgrade Recipe
Basic	Orange	1x	Starting tier — crafted directly
Advanced	Red	2x	Basic + osmium + redstone
Elite	Dark Red	4x	Advanced + diamond + obsidian
Ultimate	Purple	8x	Elite + refined obsidian + alloy

Upgrade by shift-right-clicking a machine with the next tier's upgrade item.

Step 1: 2x Ore Processing

Your first Mekanism setup:

1. **Heat Generator** — place lava source below it for early power
2. **Metallurgic Infuser** — combine osmium with carbon (coal/charcoal) to make **steel ingots**
3. **Enrichment Chamber** — 1 ore → 2 dust. Smelt dust → 2 ingots. This is 2x processing

Chain: Mine ore → Enrichment Chamber → 2 dust → Furnace → 2 ingots

Osmium processing: Mine raw osmium → Enrichment Chamber → osmium dust → Furnace → osmium ingots. Then combine osmium ingots + carbon in the Metallurgic Infuser to make steel.

Advanced Processing Chains

Mekanism offers multiple ore processing multipliers:

Multiplier	Machines needed	Process
2x	Enrichment Chamber	Ore → 2 dust → smelt → 2 ingots
3x	Enrichment Chamber + Crusher	Ore → dust → dirty dust → 2x Enrichment → 3 dust → smelt → 3 ingots
5x	Full chemical line (requires gas infrastructure)	Chemical Dissolution Chamber (sulfuric acid) → Washer (water) → Crystallizer → 5 dust → smelt → 5 ingots

Practical early setup: Start with 2x using just an Enrichment Chamber. This is good enough for early game. Move to 5x processing when you have power to spare — the chemical line needs significant Joules to run.

The 3x chain step by step:

1. Enrichment Chamber: ore → dust
2. Crusher: dust → dirty dust
3. Enrichment Chamber: dirty dust → dust (second pass = 3x total)
4. Furnace: dust → ingots

Gas Infrastructure

Mekanism adds gases as a resource type. You need them for high-tier processing:

Gas	Produced by	Used for
Hydrogen	Electrolytic Separator (split water)	Power generation, processing
Oxygen	Electrolytic Separator (split water)	Chemical processing
Sulfuric Acid	Chemical Infuser (sulfur + oxygen)	5x ore processing
Deuterium	Electrolytic Separator (heavy water)	Fusion reactor fuel
Tritium	Chemical Infuser (lithium + deuterium)	Fusion reactor fuel

Gases move through pressurized tubes. Craft **mechanical pipes** (basic) or **pressurized tubes** (for gas). Connect a **Rotary Condensentrator** to convert gas ↔ liquid for storage in standard tanks.

Digital Miner

The digital miner is Mekanism's automated mining machine. It's a mid-game goal.

How to use:

1. Craft the **Digital Miner** and place it
2. Right-click to open its GUI
3. Set a **radius** (default 32 blocks, increase up to 64)
4. Set a **filter** — by default it mines everything. Add whitelist/blacklist filters for specific ores
5. Supply power — the miner needs significant energy (4x the normal rate)
6. Press "Start" — it scans the area and begins mining

Tip: Set the miner to only mine ores (add an ore dictionary filter). This prevents it from filling with stone.

Fusion Reactor (Endgame)

The fusion reactor is Mekanism's endgame power source. It produces millions of Joules per tick — enough to power your entire base, charge a MekaSuit, and run every machine simultaneously.

Fuel production chain:

1. Pump water into an **Electrolytic Separator** → produces hydrogen + oxygen
2. Process hydrogen through a series of machines to produce **deuterium** (check JEI for the exact chain — involves Chemical Infuser with specific catalysts)
3. Combine deuterium + lithium in a **Chemical Infuser** → **tritium**
4. Feed both deuterium and tritium into the **Fusion Reactor** using pressurized tubes
5. The reactor generates power once it reaches ignition temperature (right-click to start)

Note: The fusion reactor is expensive to build and requires significant infrastructure. It's a Phase 3 project. Work up through Mekanism's generator tiers (heat → gas-burning → bio-generator) first.

AE2 — Applied Energistics 2

AE2 is about **digital item storage** and **autocrafting**. Instead of chests full of items, everything lives in an electronic network that you access from a single terminal.

Finding Certus Quartz

Certus quartz generates in two ways:

- **Meteorites** — Look for circular depressions on the surface. Dig down to find a meteorite containing certus quartz blocks and a **sky stone chest** with free AE2 components
- **Ore veins** — Certus quartz ore generates underground. Mine it with a pickaxe

Processing certus quartz:

1. Certus quartz block → mine → certus quartz item
2. Craft into **certus quartz dust** (use a crafting table or macerator)
3. Smelt dust → **certus quartz crystal**
4. Charge crystals in an **AE2 Charger** (requires power) → **charged certus quartz**

The Channel System

AE2 has a **channel system** that limits how many devices can connect through a single cable. This is the hardest concept for new players.

Cable type	Channels carried	Visual indicator
ME Glass Cable	8	No indicator
ME Smart Cable	8	Colored lines show channel usage (1-8)
ME Dense Smart Cable	32	Thicker cable, colored lines
ME Cover Cable	8	Can be hidden in walls

Channel rules:

- Each cable can carry 8 channels (32 for dense)
- Each ME Controller face provides 32 channels
- Each device (drive, terminal, bus) consumes 1 channel
- If you exceed channels, devices show "no power" (colorless cables)

Managing channels: Use **ME Smart Cable** everywhere — the colored lines instantly show you how many channels you're using (yellow = 1, green = 2, blue = 4, etc.).

Your First Network

Step 1 — Set up power: AE2 devices need power (not Create or Mek power — AE2's own power system). Build a **vibration chamber** or **AE2 energy acceptor** and connect any power source (Create alternator or Mekanism generator).

Step 2 — Build the controller: Craft an **ME Controller**. Place it. Connect power. Each face provides 32 channels — for your first network, one controller is enough.

Step 3 — Add storage:

1. Craft an **ME Drive** (holds up to 10 storage cells)
2. Craft **1k storage cells** (charged certus quartz + redstone + glass)
3. Place the drive next to the controller or connect via cable
4. Put cells in the drive

Step 4 — Add a terminal: Craft an **ME Terminal** (or **ME Crafting Terminal** — essential, lets you craft from stored items). Place it anywhere on the network. Right-click to open — all your digital items in one screen.

Connecting to the World

Device	What it does	Channel use
ME Import Bus	Pulls items from adjacent inventory into the ME network	1
ME Export Bus	Pushes items from the network into adjacent inventory	1
Storage Bus	Treats an adjacent inventory as part of the ME network (no import/export needed)	1
ME Interface	Exports items to the network AND accepts items for autocrafting	1

Practical setup:

- Place an **Import Bus** on your furnace → smelted items go directly into ME storage
- Place a **Storage Bus** on a Sophisticated Storage barrel → the barrel's contents appear in your ME terminal
- Place an **Export Bus** on your Mekanism crusher → items flow from ME storage into the machine

Autocrafting

Once your network is running, set up autocrafting:

1. Craft **Patterns** (blank pattern + items in a crafting grid)
2. Put patterns in an **ME Pattern Provider**
3. Place the provider next to an **ME Molecular Assembler** (or use cables to connect them)
4. Request the crafted item from your terminal — AE2 pulls ingredients from storage, sends them to the assembler, and returns the result

For Mekanism/Create machines: Use an **ME Interface** instead of pattern provider. Put the pattern in the interface, and the interface exports ingredients to the adjacent machine and accepts the result back.

Subnetworks (Advanced)

When your network grows beyond 32 channels, build a **subnetwork**:

1. Place a second **ME Controller** somewhere
2. Connect it to the main network via **ME P2P Tunnel** (or use **ME Quantum Link** for wireless)
3. The second controller provides 32 more channels
4. Run **ME Dense Smart Cable** between controllers

Alternative: Use **P2P tunnels** to compress 32 channels through a single cable.

AE2 Keybindings

Keybind	Default	What it does
Open terminal	Right-click on terminal	Access all stored items
Open crafting terminal	Right-click on crafting terminal	Access + craft from storage
View network status	N/A	Right-click controller with a network tool

Mid-Game Integration

By mid-game, you should have all three mods working together:

Layer	Mod	What's happening
Power	Create	Water wheels → windmills → steam engines power the base
Power supplement	Mekanism	Heat generators → gas generators supplement Create power
Ore processing	Mekanism	3x processing chain (Enrichment Chamber + Crusher)
Component crafting	Create	Mechanical press + deployers automate component crafting
Storage	AE2	Import buses on all machines pull items into the ME network
Autocrafting	AE2	Pattern providers + molecular assemblers craft on demand

Layer	Mod	What's happening
Power bridge	Crafts & Additions	Electric motors/alternators convert kinetic SU ↔ FE (Mek/Stellaris)
Rail transport	Steam 'n' Rails	Expanded train system — new tracks, semaphores, coupling blocks
Transport	Create	Conveyor belts move items between processing lines
Oil processing	TFMG	Distillation Tower → diesel, gasoline, LPG. Crack heavy fractions
Diesel power	TFMG	Diesel engines produce 2,000–16,000 SU from refined oil
Steel production	TFMG	Multi-block Blast Furnace produces steel in bulk
Ore finding	Skniro's Ores	New ore types generate in the Nether and End for mid-game expansion

Nether & End Ores (Skniro's)

Skniro's Nether and End Ores adds new ore types to both dimensions. These ores are processed through Mekanism machines and feed into Create/Mek material chains.

Nether ores: Generate in the Nether at various Y-levels. Check JEI ([R](#)) to see exactly which ores are added and their required mining levels. Most can be processed through Mekanism's Enrichment Chamber for 2x output.

End ores: Generate on the End islands after defeating the Ender Dragon. These are higher-tier materials used in advanced Mekanism machines and Create components.

Tip: Use the Explorer's Compass (Wave 2) to find specific biomes where certain ores generate. Some ores are biome-specific.

Mekanism TFMG Compatibility — Recipe Resolution

The **Mekanism TFMG Compatibility** mod resolves overlapping recipes between Mekanism and TFMG. Both mods add similar processes (steel production, oil processing, plastic/rubber), and this compatibility layer ensures they work together rather than redundantly.

What changes:

Feature	Resolution
Steel production	Mekanism's steel recipe (Metallurgic Infuser) is disabled. Use TFMG's Blast Furnace instead — it's the intended steel source for this pack
Oil processing	Both mods' oil types are unified. TFMG's Distillation Tower is the primary oil processor; Mekanism's oil chain feeds into it
Plastic / Rubber	TFMG's plastic/rubber recipes take priority. Mekanism's polyethylene chain is disabled to avoid duplicate materials
Circuitry	Mekanism's basic control circuit recipe uses TFMG's aluminum where applicable

Config: All toggles are in `config/mekanismtfgmcompat-common.toml`. Default settings are correct for this pack — no changes needed unless you want to re-enable a specific Mekanism recipe.

Late Game

- **AE2 autocrafting:** Set up pattern providers + molecular assemblers. Request any item, AE2 crafts it automatically from raw materials stored in the ME network
- **Mekanism digital miner:** Automated mining — place it, set radius and filters, it mines everything. Feed output directly into AE2 via import bus
- **Create trains:** Build tracks between bases. Assemble a train (engine + carriages). The train follows tracks automatically — use train stations to load/unload at each stop
- **Mekanism fusion reactor:** Endgame power. Requires deuterium + tritium from processed water

Endgame

- **MekaSuit:** Mekanism's endgame armor. Full flight, extreme damage resistance, fire/lava immunity, underwater breathing, auto-feed. Powered wirelessly by your base's energy grid. This is the power fantasy goal
- **AE2 autocrafting CPU:** Complex multi-step autocrafting chains. Request advanced alloys and Mekanism components — AE2 crafts them from raw materials
- **Create mega-factory:** Massive conveyor networks, train stations at every mining outpost, automated farms for Mekanism biofuel. Build the factory you've always wanted

Session Plan — Wave 1

Wave 1 is the largest and takes the most sessions. Pace yourself — each session adds one new capability.

Session	Focus	What to Build
Session 1 — Create basics	Mine andesite. Craft hand crank + millstone. Build a water wheel. Connect millstone → grind wheat and ore	Power generation (256 SU), basic ore doubling
Session 2 — Mekanism start	Find osmium. Build Heat Generator + Enrichment Chamber + Metallurgic Infuser. 2x ore processing. Make steel	Basic Mekanism line, steel for future projects
Session 3 — AE2 discovery	Find a certus quartz meteorite. Mine it. Craft certus dust → smelt → charge. Build AE2 Charger	First certus quartz, charge infrastructure
Session 4 — AE2 network	Build ME Controller + ME Drive + 1k storage cells + ME Crafting Terminal. Place a Storage Bus on your Sophisticated Storage barrel	Your first digital storage network
Session 5 — Mid-game Create	Find zinc. Make brass. Build Deployer + Sequenced Crafter + Mechanical Arm. Automate a simple recipe	Component automation, brass-tier machines
Session 6 — Mekanism expansion	Add Crusher for 3x ore. Build Electrolytic Separator for gas processing. Upgrade machines to Advanced tier	3x ore multiplication, gas infrastructure

Session	Focus	What to Build
Session 7 — Integration	Place Import Bus on Enrichment Chamber → items flow into AE2. Place Export Bus to supply Create machines. Connect Create alternator to Mekanism power	Cross-mod automation bridge
Session 8+ — Scaling up	Build Digital Miner. Expand AE2 drives. Add Create train between base and outpost. Work toward 5x ore processing	Automated mining, bulk transport
Session 9 — TFMG start	Find oil (desert/plains surface pools). Craft Pumpjack + Distillation Tower. Build Small Diesel Engine. Process oil into diesel fuel	Oil refining online, massive SU upgrade
Session 10 — TFMG expansion	Build the multi-block Steel Blast Furnace. Mass-produce steel. Find bauxite for aluminum. Upgrade to Large Diesel Engine	Industrial steel, aluminum processing
Session 11+ — TFMG + Mek integration	Wire diesel engines to power Mekanism processing. Use TFMG steel for Mekanism machines. Connect Mekanism TFMG Compatibility configs	Cross-mod fuel-to-processing pipeline

Pacing tip: You don't need to finish Wave 1 before starting Wave 1.5 (colony) or Wave 2 (exploration). Dip into exploration between factory sessions. The colony takes real-world days to grow, so start it early. TFMG oil is a mid-game addition — start it after you have brass Create components and basic Mekanism processing.

Wave 1.5 — Colony Management

Your colony grows alongside your factory. MineColonies provides autonomous NPC workers that mine, farm, craft, and guard your base. Chunk Loaders (Wave 0) keep the colony running across dimensions.

Install

See [MODLIST-V3.md](#). These mods are listed in Wave 1's Colony Automation and Chunk Loading sections.

MineColonies — Autonomous NPC Colony

MineColonies lets you build a functioning town of NPC workers. Each worker has a dedicated hut that you place, and they perform jobs autonomously — mining, farming, crafting, building, guarding.

Getting Started

1. **Craft the Town Hall** (check JEI) and place it on flat ground
2. **Right-click the Town Hall** — it opens the colony GUI
3. **Hire a Builder** — the Town Hall UI has a "Citizens" tab. Recruit a builder
4. **Place a Builder's Hut** near the Town Hall. The builder walks to it and claims it
5. **Supply materials** — put wood, cobblestone, and tools in the builder's chest (inside the hut)
6. The builder starts constructing buildings from blueprints

Core Worker Huts (build in this order)

Hut	Resource	What it does
Mine	Coal, iron, stone	Digs a shaft mine. Supplies your base with basic ores
Lumberjack	Wood	Chops trees and replants saplings. Infinite wood
Farmer	Wheat, carrots, potatoes	Plants, grows, harvests. Feeds your colony
Guard Tower	Defense	Guards patrol your base perimeter. Add more towers for larger bases
Crafter	Recipes	Crafts items from materials in storage. Set recipes in the crafter GUI

Research Tree

Open the Town Hall → Research tab. Research points accumulate from worker activities. Key gates:

Research	Colony Level	Points	Unlocks
Basic Industry	2	2000	Better mine shafts, furnace automation
Advanced Smithing	3	4000	Better tools for workers
Industrial Revolution	3	5000	Mekanism Atomic Alloy recipe
Logistics	4	8000	Warehouse, courier delivery routes

Tech gating: The Atomic Alloy required for Mekanism's ultimate-tier machines can be set to require "Industrial Revolution" research. This ensures the player can't skip from Phase 1 straight to Phase 3 tech — the colony must grow alongside the factory.

Guard Scaling

The colony's defense scales with size:

- 1 Guard Tower → covers a 20-block radius
- Each additional tower adds coverage
- Guard equipment upgrades with your colony level (leather → iron → diamond)
- Guards from multiple towers patrol overlapping areas, creating a secure perimeter

Integrating With Your Tech

- **AE2 storage bus** on the colony warehouse chest → all colony resources accessible from your ME terminal
- **Create conveyor belts** can feed materials into the builder's chest for automated restocking
- **Farmer's Delight meals** can be supplied to the colony kitchen for worker happiness bonuses
- **Chunk Loaders** keep the colony running when you're exploring other dimensions

Configuration — Wave 1.5

MineColonies — no config changes needed initially. Default colony settings (3-block worker range, normal build speed, 1-day child growth) are balanced. The mod's internal progression (builder→miner→farmer→crafter→guard, then research tree) naturally gates itself — you can't build a level 3 mine until you've leveled the builder and accumulated resources.

Session Plan — Wave 1.5

The colony takes real-world time to grow. Sessions here are check-ins between other activities.

Session	Focus	What to Do
Session 1 — Foundation	Craft Town Hall, place it, hire Builder. Place Builder's Hut, supply materials (wood, cobble, tools). Start the first build	Colony established, first building under construction
Session 2 — First workers	Place Mine hut + Lumberjack hut. Supply tools and materials. Let them work while you explore or build factory	Passive resource income (wood, coal, iron)
Session 3 — Farm + Guard	Place Farmer hut for food. Place Guard Tower for perimeter defense. Upgrade buildings to level 2	Colony self-sufficient, base defended
Session 4 — Research	Check Town Hall → Research tab. Spend accumulated research points on Basic Industry. Place Crafter hut	First research unlocked, autocrafting in colony
Ongoing	Check in every few sessions. Upgrade buildings, unlock more research, expand guard coverage. By Phase 2 your colony should be level 3 with Industrial Revolution research	Tech gate unlocked (Atomic Alloy), automated resource supply

Tip: The colony is a long-term investment. Start it early (Session 1 of Wave 1.5), check on it periodically, and let it grow in the background. By the time you need Atomic Alloy for Mekanium, it'll be ready.

Wave 2 — Exploration (World & Dimensions)

The world is larger, richer, and full of things to discover. Tectonic makes the terrain dramatic, Terralith adds overhauled biomes, YUNG's structures overhaul every vanilla structure, and three new dimensions open up at your own pace.

Install

See [MODLIST-V3.md](#). Install all mods at once.

Overworld

The Overworld is dramatically transformed. Every biome, structure, and cave system is overhauled.

Terrain Overhaul — Terralith + Tectonic

Aspect	What changed	What to look for
Mountains	Tectonic makes them taller and steeper. Terralith adds painted mountains (orange/white layers) and volcanic peaks	Build a base in a mountain valley with dramatic views
Oceans	Deeper and more varied — trench-like drops offshore	Use ships/boats to cross, lookout for ocean monuments
Biomes	Terralith adds 80+ overhauled biomes — redwood forests (200+ block trees), cherry groves, floating islands, painted mountains, and more	Cherry groves for vibes. Redwood forests for massive wood supply
Caves	YUNG's Better Caves expands cave systems — wider tunnels, deeper chasms, more open spaces	Caves are easier to navigate and more impressive. Bring torches
Rivers & Valleys	Tectonic carves deeper river valleys. Terralith adds shattered savannahs (mesa-like)	Follow rivers for scenic travel. Build bridges with Create

Pro tip: Each of these worldgen mods works best in **newly generated chunks**. Load up a world and explore in any direction. The terrain variety means every direction has something different.

YUNG's Overhauled Structures

Every vanilla structure type is expanded:

Structure	What's new	Loot expectation
Better Dungeons	Multi-room dungeons with spawners, traps, and treasure rooms	Good early loot — iron, gold, enchanted books
Better Mineshafts	Larger, multi-level mineshafts with support beams and loot caches	Rail, minecarts, ore samples
Better Strongholds	End portal room is deeper. More libraries, prisons, and corridors	Enchanted books from libraries. Good mid-game exploration
Better Desert Temples	Multi-room underground temple. Traps, puzzles, treasure vault	High-value loot in the vault room. Watch for pressure plates
Better Jungle Temples	Vertical temple with multiple challenge rooms. Redstone puzzles	Unique loot from solving puzzles. Opening the vault disarms the final trap
Better Ocean Monuments	Redesigned with new rooms and challenges. More elder guardians	Better sponge room loot, treasure blocks
Better Witch Huts	Expanded hut on stilts with potion lab underneath	Brewing supplies, potions, redstone
Better Nether Fortresses	Larger, with distinct districts (blaze, wither skeleton, strider). Bridges connect sections	Nether wart, blaze rods, wither skeleton skulls

Structory + Towns & Towers

Dozens of hand-crafted structures spawn across the Overworld. Some are small (camps, shrines), others are large (castles, towers, walled towns). Villages generate with unique architecture from Towns & Towers — each biome style gets its own building designs.

What to do: Explore! Mark interesting structures on your map (Xaero's: add waypoints with **Ctrl+Left Click** on the fullscreen map). Return to promising ones with an Explorer's Compass.

Explorer's Compass — Navigation Tool

The Explorer's Compass is essential for finding specific biomes and structures:

- 1. **Craft:** 1 compass + 4 iron ingots in a + pattern
- 2. **Right-click** to open the GUI
- 3. **Search tab:** Type a biome or structure name to filter. Supports partial matching — "cherry" finds all biomes with "cherry" in the name
- 4. **Select an entry:** The compass immediately points toward the nearest instance of that biome/structure. Distance in blocks is shown
- 5. **Follow the arrow:** Hold the compass in your main hand and walk. The arrow updates in real-time
- 6. **Waypoint sync:** When you find the target, add a waypoint in Xaero's map (fullscreen map → Ctrl+Left Click)

Keybinding note: The compass does not have its own keybind by default. Craft it, hold it, right-click. If right-click does nothing, check Controls → Explorer's Compass for a conflicting keybind.

Immersive Aircraft — Flyable Vehicles

Immersive Aircraft adds vanilla-style aircraft that let you fly across the Overworld and between dimensions. Available in mid-game (requires iron and basic Create/Mek materials).

Aircraft Types

Vehicle	Fuel	Seats	Cargo	Best for
Biplane	Coal/charcoal	1	None	Fast personal flight, combat strafing
Airship	Coal/charcoal	3	9 slots	Exploration with friends, cargo runs
Cargo Airship	Coal/charcoal	1	27 slots	Transporting materials between bases
Warship	Coal/charcoal	3	9 slots	Armed combat, colony defense support
Gyrocopter	Coal/charcoal	1	None	Agile short-range flight, tight spaces

How to Use

- 1. **Craft** an aircraft at a crafting table (check JEI — each requires iron, wool/leather, and basic components)
- 2. **Place** it on the ground — aircraft are placed blocks, not items

3. **Right-click** to enter. **W** to ascend, **S** to descend, **A/D** to rotate
4. **Fuel**: Any furnace fuel (coal, charcoal, lava buckets) goes in the aircraft's fuel slot
5. **Repair**: Craft with iron ingots to repair damage. Mobs can damage parked aircraft

Keybinding: Aircraft controls are bound to movement keys by default. Check Controls → Immersive Aircraft for custom bindings.

Progression Notes

- Aircraft are **mid-game** (Phase 2) — recipes require iron and basic Mechanism/Create components. They're not available on day one.
 - An **airship** makes exploring Terralith's vast biomes, spotting YUNG'd structures, and reaching dimension portals dramatically faster.
 - Aircraft work in the **Aether** — fly between sky islands without building bridges.
 - Aircraft can be damaged by mobs and environmental hazards. Park them inside or on a high platform.
 - **No config changes needed** — default fuel consumption and durability are balanced for the pack.
-

The Aether — Sky Dimension

A sky dimension floating above the clouds. It's safe, beautiful, and offers a gentle adventure curve.

Portal Building

1. Gather **10-14 glowstone blocks** (found in the Nether, or from wandering traders)
2. Build a rectangular frame (4×5 minimum, like a Nether portal) using glowstone
3. Fill the inside with a **water bucket** (right-click with water inside the frame — one bucket is enough; unlike the Nether, it doesn't consume the source)
4. Step through. You'll ascend into the Aether

Troubleshooting: If the portal doesn't activate, check:

- The frame must be at least 4 blocks wide × 5 blocks tall (interior)
- Glowstone blocks only (not glowstone lamps or other light blocks)
- You must fill the frame with a water source block, not flowing water

What to Do

Phase 1 — Arrival: You spawn on a sky island. Your tools and weapons work fine — bring them from the Overworld. Explore surrounding islands. Key early resources:

- **Ambrosium** — Aether's coal equivalent. Shoots light, powers items
- **Zanite** — Aether's iron equivalent. Mining level increases with durability loss (weaker zanite tools mine higher tiers)
- **Aerwhales** — Passive floating whales. Fleece drops for decorative blocks

Phase 2 — Gold Dungeon: Find a bronze dungeon (tower with bronze block roof). Defeat the **Slider** (a moving block boss — hit it when it's blue, avoid when red). Loot the bronze treasure chest for **Golden Amber** and access to the next tier.

Phase 3 — Silver Dungeon: Deeper in the Aether. More dangerous mobs. Defeat the **Valkyrie Queen** for the **Valkyrie tools** — flight-enabled melee weapons. Key loot:

- **Gravitite** — floats upward when right-clicked, falls when sneak-right-clicked. Used for the best Aether gear
- **Valkyrie Lance** — long-reach throwing weapon
- **Aechor Petal** — poison source for arrows

Phase 4 — Endgame: The final Aether boss requires exploring the highest crystal islands. The **Sun Spirit** is the ultimate challenge — bring fire resistance.

Tip: The Aether isn't required for any other dimension or tech progression. Visit when you want a change of pace. Your Mekanism gear works fine here — with the MekaSuit, you can trivialize all Aether combat.

Twilight Forest — Progression Dimension

A twilight-dimension with a strict boss progression system. Each boss kill unlocks the next area. The dimension is dark, enchanted, and full of maze-like structures.

Portal Building

1. Find a 2×2 pool of water (any depth, natural or player-dug)
2. Surround it with flowers (12 total — one per block on the perimeter of the 2×2. Any flowers work: dandelions, poppies, tulips, etc.)
3. Throw a **diamond** into the water pool
4. Lightning strikes the pool. Step into the water to descend into the Twilight Forest

Note: The portal is one-way down. To return, craft a **Twilight Portal** block or use a Waystone (place one in the Twilight Forest as soon as you arrive).

Boss Progression

The dimension locks areas behind boss kills. A status screen (accessible from your inventory) shows which bosses you've defeated and which areas are unlocked:

Order	Boss	Location	Strategy	Key item unlock
1	Naga	Naga Courtyard (stone arena in the forest)	Melee boss — strafe its charges, hit the tail segment	Access to Lich Tower
2	Lich	Lich Tower (tall stone tower)	Ranged fight — dodge projectiles, hit during cooldown. His shield reflects projectiles	Naga/Lich loot opens Twilight Swamp
3	Minoshroom	Labyrinth (underground mushroom maze)	Mob in a maze — find the minotaur room, kill it, grab the Meef Stroganoff	Meef Stroganoff is a food item. Opens Firefly Forest

Order	Boss	Location	Strategy	Key item unlock
4	Hydra	Hydra Lair (swamp arena, three heads)	Three-headed fire-breathing dragon. Dodge fireballs, melee the heads. Hydra drops Fiery Tears	Fiery Tears enable Ur-Ghast fight
5	Ur-Ghast	Dark Tower (tall dark brick tower in Dark Forest)	Flying ghost boss in a dark tower. Shoot its tentacles, then the main body. Dark Forest has a blindness effect — bring milk	Opens Snowy Forest + Highlands
6	Snow Queen	Aurora Palace (ice castle in the Snowy Forest)	Ice-themed arena with pillars. Push her into pillars to break her shield. Avoid ice arrows	Opens Final Castle
7	Final Castle	Castle (Highlands)	Multi-boss fight in the final castle. Requires weapons from all prior bosses	Endgame Twilight loot, includes the Giant's Pick (diamond-tier, 5×3 mining area)

Key Items to Craft/Find

- **Ironwood** gear — smelt ironwood ingots from iron + magic beans. Good starter armor
- **Fiery** gear — craft with fiery ingots (iron + blaze powder). Good mid-tier weapons
- **Knightmetal** — from the Lich Tower. Decent armor
- **Phantom** armor — allows creative flight for short bursts. Crafted from phantom membranes dropped by Ur-Ghast's tentacles
- **Giant's Pick** — mines a 5×3×5 area. From the Final Castle

Navigation Tips

- The Twilight Forest is **always twilight** — no sunlight, no day/night cycle. Mobs can spawn any time
- Place **Waystones** frequently — the dimension is large and you'll revisit areas
- The **Twilight Forest Maze Map** (crafted from paper + torch) reveals the layout of mazes and castles
- Bring **milk buckets** for the Dark Forest blindness effect
- Your AE2 wireless terminal, Mekanism gear, and Create tools all work in the dimension

Deeper and Darker — The Otherside

A deep-dark-themed dimension accessed through the Deep Dark. It's ominous, quiet, and filled with sculk.

How to Enter

1. Find a **Deep Dark** biome in the Overworld (deep underground, usually at Y=-40 or below)
2. Locate an **Ancient City** (use the Explorer's Compass — search "ancient city" structure)
3. Find and kill the **Warden** (the blind, sound-seeking boss of the Deep Dark). It drops a **Warden Heart**
4. Craft: Warden Heart + 8 sculk blocks → **Otherside Portal Catalyst** (check JEI for exact recipe)

5. Build a frame of **reinforced deepslate** (at least 4×5 interior). Activate it by right-clicking the portal catalyst on the frame
6. Step through

Warden strategy: The Warden detects by sound. Use wool blocks to silence your footsteps. Sneak near sculk shriekers (don't trigger them). Throw snowballs or projectiles to distract it. Hit and run — it has 500 HP but slow attacks.

What to Expect

- **4 new biomes** — sculk forests, frozen deep, ancient cities expanded, new cave types
- **8 new mobs** — sculk-infected variants, stalker (ranged attacker), shriek worm (ambush), and the **Sculk Leech** (heals when dealing damage)
- **Miniboss:** The **Shattered Warden** — stronger Warden variant in the deepest areas. Drops **Sculk Transmitter** components
- **Key loot: Sculk Transmitter** — an Elytra upgrade that allows teleportation mid-flight. Combine with an Elytra at a smithing table

Gear: Warden-themed armor and tools are available. They're roughly equal to diamond but with unique abilities (sound-based attacks, silence fields). A viable alternative tech path if you're tired of strip-mining for diamonds.

Difficulty: Harder than the Aether, easier than Twilight Forest's Dark Forest. Come with at least iron-tier armor and a decent weapon. Bring blocks to pillar up and escape corners.

End Overhaul

After you defeat the Ender Dragon, the End becomes a full dimension worth exploring.

What Changes

BetterEnd NeoForge adds 24+ new biomes to the End, transforming it from a barren void into a diverse alien landscape:

Biome type	Description	Notable resources
Floating islands	Large islands with unique vegetation. End lakes, end moss	New wood types, building blocks
Crystal caves	Underground caverns with glowing crystals	Crystal materials for tools/gear
Mushroom fields	Giant end-themed mushrooms	Magic-related crafting materials
Ice biomes	Frozen wastelands on the edge of the End	—
Chorus forests	Dense chorus plant forests	Chorus flowers, new food

Moog's End Structures adds towers, ruins, and dungeons across the End — many with loot better than vanilla End cities.

Ender's Delight adds food that can be cooked from End ingredients — gives useful status effects for End exploration.

Progression

Phase	Activity	Reward
Post-Dragon	Explore the outer End islands. BetterEnd biomes spawn here	New materials, tier-appropriate gear
Mid-End	Tackle Moog's structures. Find hidden rooms and bosses	Unique weapons and loot
Endgame End	Find the rarest biomes and build an End base	All materials needed for top-tier End gear

Tip: Set up a Waystone in the End as soon as you arrive. Build a small AE2 subnet with a wireless terminal to access your main base storage while exploring.

Session Plan — Wave 2

Exploration is something you dip into between factory sessions. No need to binge it.

Session	Focus	What to Do
Session 1 — Scout	Craft Explorer's Compass. Search for a cherry grove or redwood forest biome. Mark waypoints on your map. Find a YUNG's dungeon and loot it	First taste of the transformed world, early loot
Session 2 — Aether visit	Gather glowstone (from the Nether or wandering traders). Build the Aether portal. Explore the first sky island, collect ambrosium and zanite	A new dimension, safe change of pace
Session 3 — Twilight Forest entry	Find a 2×2 water pool, surround with flowers, throw in a diamond. Enter the Twilight Forest. Place a Waystone immediately. Defeat the Naga	First Twilight Forest boss, door to progression
Session 4+ — Deep exploration	Work through Twilight Forest bosses at your own pace. Visit Deeper and Darker (requires finding an Ancient City). Explore the End after killing the dragon	Boss loot, dimension-specific resources

Pacing tip: Explore between factory builds. When you're waiting for Mekanism to process or Create to assemble, grab your compass and walk in a random direction. Every direction has something new.

Wave 3 — Equipment Magic & RPG

Every weapon can roll with random affixes, every accessory provides a unique ability, and a skill tree gives every action meaning. This wave makes you **feel** your character grow alongside your tech progression.

Install

See [MODLIST-V3.md](#). Install all at once.

Skill Tree (RPG Series) — Persistent Character Growth

Every action in the game earns XP. As you level up, you earn skill points to spend on permanent stat bonuses. This works alongside everything else — mining ore, killing mobs, crafting items, cooking food, exploring new biomes.

Getting Started

1. **Set the keybind:** Controls → Skill Tree → set a key you'll remember (e.g., **P** for "powers")
2. Press the key to open the skill tree GUI
3. You'll see attributes organized in a tree layout

Attribute Categories

Attribute	Effect	Who benefits
Strength	+melee damage, +mining speed	Everyone — more damage and faster mining
Vitality	+max HP, +natural regen	All players — survival is always useful
Agility	+movement speed, +attack speed	Combat-focused players, kiting bosses
Magic	+spell damage, +mana regen	Less useful without a dedicated spell mod, but affects Ars Nouveau if installed
Defense	+armor toughness, +damage reduction	Tank playstyle

How XP works: Every ore you mine, every mob you kill, every item you craft, every crop you harvest generates XP. The amount scales with the action's complexity. Smelting 100 iron ingots gives more XP than smelting 1.

Dependency note: **Pufferfish's Skills** is the underlying framework that powers the Skill Tree. It handles XP tracking, attribute modification, and skill point allocation behind the scenes. Install it alongside the Skill Tree mod — no separate configuration or interaction needed.

Tip: The skill tree is lowest-investment, highest-return mod in this wave. Set the keybind and start earning XP from minute 1.

Simply Swords — Unique Weapons

60+ unique weapons with passive and activated abilities. Each weapon drops from mobs (~2% drop rate) and appears in chests. No two feel the same.

Weapon Categories

Category	Examples	Playstyle
Heavy	Greathammer, Greatsword, Glaive	Slow, high damage, knockback. Two-handed. Best for tanky builds
Medium	Longsword, Katana, Claymore	Balanced speed and damage. One-handed or two-handed. Versatile
Light	Rapier, Dagger, Sai, Cutlass	Fast attack speed, low damage per hit. Armor penetration on rapiers. Stacking debuffs on daggers
Polearms	Spear, Halberd, Pike	Extended reach. Can hit from 2-3 blocks away. Throwable (spears)
Special	Scythe (AoE sweep), Soul Harvester (soul-steal)	Unique mechanics — scythes hit all mobs in an arc

How to Find Weapons

- **Mob drops:** Every hostile mob has ~2% chance to drop a Simply Swords weapon. Zombies drop common, skeletons drop ranged-adjacent, creepers drop explosives-adjacent
- **Chests:** Dungeon chests, mineshaft chests, YUNG's structure chests all have higher roll chances
- **Trading:** Some weapons can be traded from weaponsmith villagers
- **Bosses:** Higher guaranteed drop chance from Twilight Forest bosses and Cataclysm bosses

Using Weapon Abilities

Hold **Ctrl** while hovering over a weapon in your inventory to see its ability description. Abilities include:

Ability	Effect	Found on
Whirlwind	Spin attack hitting all nearby enemies	Greathammer, Greatsword
Impale	Hold to charge, release for a piercing thrust	Spear, Pike
Backstab	+300% damage when attacking from behind	Dagger, Sai
Lifesteal	Heal for 10% of damage dealt	Soul Harvester, Scythe
Execute	+50% damage to targets below 30% HP	Axe variants
Slowness	On-hit slow effect for 3 seconds	Ice-themed weapons

Tip: The **Explorer's Compass** can find structures with higher weapon chest rates. Set up a weapon hunt route: find a YUNG's dungeon → loot it → compass to the next one.

Apotheosis — Loot & Enchanting Overhaul

Apotheosis transforms every gear drop and enchantment in the game. This is the single largest change to how gear feels.

Rarity System

Every item that drops can roll with a rarity:

Rarity	Affix slots	Identifying color	Drop source
Common	0	White	Basic mobs, mining
Uncommon	1	Yellow	Hostile mobs, basic chests
Rare	2	Cyan	Dungeon chests, mini-bosses
Epic	3	Purple	Bosses, late-game chests
Mythic	4	Gold/Magical	Endgame bosses, Cataclysm bosses

Higher rarities have more **gem sockets** and roll better **affixes** (bonus stats).

Affixes — Random Stats on Gear

Every armor piece, weapon, and tool can roll with bonus stats. Examples:

Affix	Effect	Gear type
Ruthless	+crit damage, +crit chance	Melee weapons
Arcane	+spell damage, +mana	All weapons
Sturdy	+armor, +knockback resistance	Armor
Nimble	+movement speed, +attack speed	Boots, leggings
Plentiful	+mining speed, +fortune	Tools
Titanic	+max HP, +natural regen	Chestplate

Tip: Check every gear drop. A Common helmet with a good affix might be better than an Epic one with useless stats. Disenchant unwanted gear at a salvaging table for gem dust.

Gem System

Gems are socketable items that add permanent stats to your gear:

Gem tier	How to get	Effect
Rough	Drops from mobs and chests	Basic stat boost
Cut	Rough gem + gem cutting station	+50% stat value. Requires gem dust
Flawless	2 cut gems of same type + gem cutting station	+100% stat value. Requires more dust

Gem cutting station: Craft a gem cutting station (check JEI). Place a rough gem in the left slot, gem dust in the center, collect a cut gem from the right. Gem dust comes from breaking unwanted cut gems in a crafting grid.

Gem socketing table: Craft a gem socketing table (check JEI). Place your item in the center, a gem in the socket slot. Each gear piece has limited sockets — expand with socketing tools.

Enchanting Overhaul

The vanilla enchanting table is replaced. New features:

- **Higher level cap:** Enchantments can go past vanilla limits. Sharpness VIII is possible
- **Better bookshelf scaling:** Bookshelves now have tiers. Higher-tier bookshelves enable higher enchantment levels
- **Library of Alexandria:** Craft one (expensive — requires diamonds, obsidian, and enchanted books). Stores all your enchanted books in one block-size inventory
- **New enchantments:** Apotheosis adds 30+ new enchantments. Check JEI for full list

Boss Affixes

Boss mobs can spawn with random modifiers displayed under their health bar:

Modifier	Effect	Strategy
Tough	+50% HP, damage reduction	Bring high-damage weapons and Lifesteal
Blinding	Blinds hit players	Wear blindness-resistant gear or bring milk
Clumsy	Disarms players on hit	Use speed — don't get hit
Plagued	Inflicts poison/wither on hit	Bring antidotes and regeneration
Huge	2× size, more knockback	Stay mobile. Don't get cornered
Chaotic	Random effects per hit	Unpredictable — bring all potions

Tip: Boss affixes are the main source of Mythic gear. Hunt for affixed bosses when you need endgame upgrades.

Relics RPG — Legendary Accessories

Relics are unique accessories that go in your **Curios slots**. You have 6 Curios slots by default: Ring (2), Amulet (1), Belt (1), Gloves (1), Back (1), Charm (1). Right-click to open your Curios inventory — default keybind set it to **Y**.

Relic	Slot	Ability	Where to find
Monkey Talisman	Charm	Double jump (press jump again mid-air)	Vanilla dungeon chests
Holy Water	Charm	Right-click to cleanse all negative effects	Desert temples, churches
Lucky Coin	Charm	+luck for better loot drops	Mineshaft chests, pillager outposts
Captain's Hook	Gloves	Right-click to pull entities toward you	Shipwrecks, ocean ruins
Sacred Wardstone	Amulet	Absorbs damage, breaks after absorbing X damage	Stronghold chests

Relic	Slot	Ability	Where to find
Magic Mirror	Back	Right-click to teleport to spawn/home point	End cities, libraries
Heart of the Void	Ring	+max HP, slow regen	Nether fortresses, bastions
Blazing Feather	Back	+15% movement speed	Jungle temples, pagodas
Drowned Belt	Belt	Underwater speed boost, +swim speed	Ocean monuments, shipwrecks

Stacking: You can wear multiple relics of different types. Two rings = two ring relics. Find duplicates to trade or salvage.

Finding relics by structure: Use the Explorer's Compass → select a YUNG's structure → loot it → compass to the next. Relics have higher drop rates in YUNG's overhauled structures than vanilla ones.

Runes — Socketable Weapon Modifiers

Runes are socketable modifications for weapons and tools. Unlike Apotheosis gems (permanent), runes are **swappable** — you can change them between fights.

How to Get Runes

- **Loot chests** in any structure
- **Craft at a Rune Altar** — place 4 cobblestone in a 2×2, top with a lapis block, right-click with a rune fragment. Check JEI for exact recipes
- **Mob drops:** Runes have a low chance to drop from any hostile mob

Rune Types

Rune	Effect	Best on
Fire Rune	On-hit fire damage (3 seconds)	Fast weapons (daggers, rapiers) for stacking burn
Ice Rune	On-hit slow (2 seconds)	Kiting weapons (spears, scythes)
Lightning Rune	Chance on-hit to strike lightning	Heavy weapons (single big hits)
Poison Rune	On-hit poison (5 seconds)	Dot builds, boss kiting
Lifesteal Rune	Heal 5% of damage dealt	Tank builds, slow weapons
Critical Rune	+10% crit chance	Medium weapons (balanced speed/crit synergy)

Using Runes

1. Craft a **Rune Pouch** (leather + string + lapis) — stores all your runes. Set a keybind for it (Controls → Rune Pouch)
2. Place your weapon and a rune at a **Rune Altar** (craft: 4 stone + 1 lapis block)
3. The rune is applied. The weapon gains the rune's effect

4. To swap: place the weapon back at the Rune Altar with a different rune

Tip: Keep 2-3 different rune loadouts in your Rune Pouch. Switch between Fire (general combat), Lifesteal (boss fights), and Ice (kiting tough mobs).

Session Plan — Wave 3

Gear upgrades happen naturally as you play. These sessions show you what to look for.

Session	Focus	What to Do
Session 1 — Skill tree	Set skill tree keybind (P recommended). Open it and spend your first points on Strength (mining speed/melee damage). This is passive — XP earns automatically from everything you do	Permanent character progression started
Session 2 — First affix loot	Check every gear drop. Look for Rare+ items with useful affixes (Ruthless for melee, Sturdy for armor). Don't salvage anything until you understand what's good	First taste of the loot system
Session 3 — Weapon hunt	Use Explorer's Compass to find a YUNG's dungeon. Loot it for Simply Swords weapons. Try out a Rapier (fast, shield-piercing) or Greathammer (AoE, knockback)	Your first unique weapon, learn its ability
Session 4 — Gem cutting	Collect rough gems from chests and mobs. Build a Gem Cutting Station. Cut your first Rough → Cut gem. Socket it into a weapon or armor piece	Permanent gear upgrade, gem system understood
Session 5 — Relic hunting	Explore different structure types for Relics RPG items. Monkey Talisman (double jump from vanilla dungeons) and Lucky Coin (better loot from mineshafts) are early priorities	Accessory slots filled, unique abilities
Session 6+ — Boss affix hunting	Search for affixed boss mobs (they have a colored name above their health bar). These are your best source of Mythic gear. Bring good weapons and food	Endgame gear progression

Tip: Wave 3 gear changes how every other wave feels. A Rare+ weapon with the right affix makes combat in Twilight Forest and Cataclysm dramatically easier. Invest in your gear before tackling tough bosses.

Wave 4 — Food, Farming & Comfort

Farmer's Delight brings a full cooking system. This wave is entirely optional — you can survive on baked potatoes forever — but cooking gives powerful status effects, better healing, and a cozy kitchen to build around.

Install

See [MODLIST-V3.md](#).

Farmer's Delight — Cooking Hub

Farmer's Delight adds new crops, a cutting board for ingredient prep, a cooking pot for multi-ingredient meals, and kitchen storage.

New Crops & Farming

Crop	Grows on	What it's for
Tomato	Farmland. Bush grows, pick when red	Soups, sandwiches, salads
Cabbage	Farmland. Full head when ready	Coleslaw, cabbage wraps, ferments
Onion	Farmland. Pick when shoots show	Savory dishes, soups
Rice	Grows in shallow water (1 block deep on dirt/mud)	Hearty meals, sushi (in addons)

Harvest With Ease (from Wave 0) works on all these crops — right-click to harvest, the plant stays for regrowth.

Wild crops: Found in the wild — look for **wild tomatoes** (plains/forests), **wild cabbage** (beaches), **wild onions** (taiga), and **wild rice** (swamps). Breaking them gives the seed item.

Cutting Board — Ingredient Prep

Craft: 1 knife + 2 planks (horizontal row)

Place the cutting board on the ground. Right-click with the cutting board in hand after placing it to open the UI, or just right-click the placed cutting board.

How it works:

- 1. Place an ingredient in the left slot
- 2. The cutting board processes it (chops, slices, peels)
- 3. Collect the output from the right slot

Essential recipes (always check JEI — many ingredients have cutting board recipes):

Input	Output	Use
Any cooked meat	Meat strips	Sandwiches, salads
Cabbage	Cabbage leaves	Wraps, coleslaw
Bread	Bread slices	Toast, sandwiches
Carrot	Carrot slices	Salads, soups
Apple	Apple slices	Desserts, snacks
Cheese (if available)	Cheese wedge	Hearty meals

Tip: A knife in your off-hand while chopping lasts longer (the cutting board uses the knife's durability). An iron knife lasts for dozens of chops.

Cooking Pot — Meal Preparation

Craft: 2 iron ingots + 2 planks + 1 bowl (check JEI for exact shape)

Place the cooking pot over a **heat source**. Valid heat sources (from best to worst):

- **Stove** — fastest cooking. Craft: 4 bricks + 4 iron nuggets
- **Campfire** — medium speed. Craft: 3 logs + 3 sticks + 1 coal
- **Lava** — slowest. The pot must sit directly above the lava source block
- **Torch/Campfire block below** — does not work. Needs a proper heat block

Cooking process:

1. Right-click the pot to open the GUI
2. Fill the **water slot** (bucket or bottle on the left side)
3. Add ingredients (up to 6) in any order
4. Wait for the progress bar to fill
5. Collect your meal — most meals give **6 servings** with bowl stacking

Meal categories (check JEI for full list):

Category	Example	Effect
Soups	Tomato soup, vegetable soup, beef stew	Good saturation, easy ingredients
Sandwiches	Chicken sandwich, egg sandwich, BLT	On-the-go eating, high saturation
Salads	Mixed salad, coleslaw, fruit salad	Refreshing, good for hot biomes
Pasta	Pasta with meat sauce, pasta with veggies	Filling, complex ingredients
Rice meals	Rice pudding, fried rice	Requires rice farming

Kitchen Storage

Craft **oak/pantry cabinets** (check JEI for recipe — varies by wood type). These function like chests but match your kitchen aesthetic. They hold as much as a chest.

Tip: Set up your kitchen near your AE2 network. On the one hand, you can cook manually. On the other, your ME drive can stock ingredients and meals.

Brewin' And Chewin' — Drinks & Fermentation

Adds alcohol brewing with aging mechanics. Grow hops, barley, and grapes on standard farmland.

Brewing Station — Fermenting Base

Craft: Check JEI (uses a barrel-like recipe)

1. Place the **brewing station** (function block, not decoration)
2. Right-click to open the GUI
3. Add **water** + your main ingredient:
 - **Barley** → beer base (wort)
 - **Grapes** → wine base (must)

- **Honey** → mead base
 - **Potatoes** → spirits base (mash)
4. Wait for fermentation to complete (the station processes the mixture)
 5. Collect the **fermented mixture** — it's now ready for aging

Aging Barrel — Flavor Development

Craft: Check JEI (wooden barrel)

1. Place the aged mixture in the **aging barrel**
2. Right-click to seal it
3. Every in-game day, the drink ages one stage:
 - **Young** → **Matured** → **Aged** → **Vintage** (4 stages)
4. Each stage increases the drink's **status effect potency and duration**
5. Break the barrel to collect the finished drink

Optimal aging: Let everything reach **Aged** (stage 3). Only push to Vintage (stage 4) for special occasions — the improvement is minor but the waiting is long.

Drink Effects

Drink	Effect at Matured	Effect at Vintage
Beer	Resistance I (2 min)	Resistance II (5 min)
Wine	Haste I (2 min)	Haste II (4 min)
Mead	Speed I (2 min)	Speed II (4 min)
Whiskey	Strength I (30 sec)	Strength II (1 min). Short but strong
Vodka	Night Vision (3 min)	Night Vision + Fire Resistance (5 min)

Practical use: Brew a batch of each, age them all to Matured/Aged, and keep them in your Curios backpack or kitchen storage. Pop a drink before boss fights for free status effects. Mead stacks with Swiftiness potions for even more speed.

Bottle Return

After drinking, you get an **empty bottle** back. Reuse it at the brewing station — no glass waste.

Storage Delight — Kitchen Furniture

Storage Delight adds decorative kitchen storage blocks that work as functional inventories:

Block	Slots	Use
Cabinet	27	Counter-height storage, matches kitchen aesthetic
Drawer	18	Slimmer, fits under counters
Shelf	9	Wall-mounted spice/ingredient display

Tip: Use cabinets and drawers in your kitchen for early-game ingredient storage before AE2 comes online. Later, place an ME Storage Bus on a cabinet to bridge it into your digital network.

More Delight — Extra Meals & Knives

More Delight adds new recipes and tools that extend Farmer's Delight:

- **Extra meals:** Sandwiches (combine bread + any filling), toasts (bread + toppings), stuffed mushrooms, and more
- **Knife variants:** Diamond and netherite knives that last much longer than iron
- **New cooking pot recipes:** Stews, soups, and pasta dishes using new ingredient combinations

All recipes are visible in JEI. No special blocks needed — everything uses the existing cutting board and cooking pot.

My Nether's Delight — Nether Food

Adds Nether-themed ingredients and recipes. Turns Nether exploration into a food-gathering expedition:

- **Nether crops:** Grow in soul sand or warped/crimson nylium
- **New ingredients:** Glowstone berries, soul berries, nether-root vegetables
- **Recipes:** Spicy stews that give Fire Resistance, soul-force snacks that give Night Vision

Tip: Set up a Nether farm for ingredients that give combat-relevant status effects. Glow berry pie gives Haste II — useful before a mining expedition.

Ocean's Delight — Seafood

Adds seafood dishes using ocean mob drops and kelp:

- **Ocean ingredients:** Squid tentacles (squid drops), fish fillets, seaweed, turtle eggs
- **Recipes:** Seafood stew, grilled fish, kelp salads, sushi rolls
- **Status effects:** Seafood gives Water Breathing and Dolphin's Grace — perfect for ocean monument raids

Tip: Build a fish farm (breed fish near a fishing spot) or farm kelp for a steady supply of ocean ingredients. Combined with My Nether's Delight, you can prepare meals for any environment.

Session Plan — Wave 4

Cooking is a side activity you build up gradually. Start small and expand as you find ingredients.

Session	Focus	What to Build
Session 1 — Kitchen tools	Craft a cutting board (knife + planks) and cooking pot (iron + planks). Build a stove (bricks + iron nuggets) as heat source. Find wild tomato, cabbage, onion, rice seeds	Functional kitchen, first ingredients
Session 2 — Farm	Plant tomato, cabbage, onion, rice near water. Use Harvest With Ease for easy harvesting. Cook your first meal (tomato soup is the easiest)	Sustainable ingredient supply

Session	Focus	What to Build
Session 3 — Brewing	Find barley, grapes, honey. Build a brewing station. Start your first batch of beer/wine/mead. Place an aging barrel	First drink fermenting, waiting for it to age
Session 4 — Kitchen expansion	Craft Storage Delight cabinets and drawers for decorative ingredient storage. Try More Delight recipes (sandwiches, toasts). Collect Nether ingredients for My Nether's Delight	Full kitchen, all FD addons in use
Ongoing	Age drinks to Matured/Aged for boss-fight buffs. Explore oceans for Ocean's Delight ingredients. Stock your AE2 system with ingredients for autocrafting meals on demand	Buff food always available for combat

Session Plan — Wave 4.5

Bounties are passive — check boards whenever you visit a village.

Session	Focus	What to Do
Any session	Visit any village. Find the bounty board (a lectern-like block). Check available quests. Complete gathering quests (deliver items from your AE2 system) for free rewards	Passive XP and loot
Ongoing	Boards regenerate quests periodically. Check back after a few play sessions. Hunting and exploration quests are good reasons to visit new biomes or fight new mobs	Continuous passive progression

Wave 4.5 — Quests

Procedural bounty boards give you optional goals as you explore. No quest book, no authoring, no pressure.

Install

See [MODLIST-V3.md](#).

Bountiful — Bounty Boards in Villages

Bountiful places **bounty boards** in every village. Right-click a board to see available quests. Each quest asks for something:

Quest Type	Example	Reward
Gathering	Deliver 16 iron ingots	Emeralds, XP, rare items
Hunting	Kill 8 creepers	Combat loot, XP
Exploration	Find a specific biome/biome type	Map items, navigation gear
Crafting	Deliver crafted items (tools, armor)	Ingots, gems

Quests are **procedurally generated** — they pull from items and mobs that actually exist in your world. No manual quest creation needed. Bountiful also has a **Decree** system that lets you blacklist certain quest types if you don't enjoy them.

How to use:

1. Find any village
2. Locate the bounty board (a lectern-like block near the center)
3. Right-click to open the quest list
4. Select a quest you can complete
5. The required items appear in the board's slots — drop them in
6. Collect your reward
7. New quests regenerate periodically

Tip: The board shows multiple quests at once. Take gathering quests for materials you already have in your AE2 system — free rewards for zero extra work.

Wave 5 — Combat & Mobs

Your powerful gear finally has something worthy to fight. This wave adds better combat mechanics, tougher mob variants, epic boss fights, and roguelike dungeon structures.

Install

See [MODLIST-V3.md](#).

Better Combat — Weapon Feel

Better Combat changes how every weapon **feels** in your hands. It replaces Minecraft's single-swing animation with weapon-specific movesets:

Weapon type	Speed	Reach	Animation	Best for
Daggers	Very fast	2 blocks	Quick stab	Debuff stacking, fast crit builds
Swords	Fast	3 blocks	Horizontal arc	General combat, crowd control
Rapiers	Fast	3.5 blocks	Thrust (ignore shields)	Shield-breaking, precision
Greatswords	Slow	4 blocks	Wide sweep	Group clearing, knockback
Axes	Medium	3 blocks	Overhead chop	Armor penetration, single target
Spears	Medium	4.5 blocks	Forward thrust	Kiting, reach advantage
Hammers	Slow	3.5 blocks	Overhead slam	Knockback, stun
Greathammers	Very slow	4 blocks	Ground pound	AoE knockback, crowd control

Config: Mod Menu → Better Combat. You can tweak:

- **Reach distance** per weapon type (longer = hit farther, easier to hit)
- **Sweep radius** (wider = hits more mobs)

- **Attack speed** scaling
- **Visual effects** (particles, trail colors)

Enchantment Descriptions

Hover over any enchantment in a tooltip (on an item, in a book, or in the enchanting table GUI) — a detailed description of what the enchantment does appears.

No config needed — install and forget.

Dependency note: **Ritchie's Projectile Library** is a library mod required by Create Big Cannons. It provides custom projectile trajectories and hit detection. Install it alongside Create Big Cannons — no separate configuration needed.

Mutant Monsters — Boss Variants

Mutant versions of vanilla mobs. Each is tougher than the original but drops unique gear. With your Epic+ Apotheosis gear, these are satisfying challenges rather than roadblocks.

Mutant	Where it spawns	What to expect	Drops
Mutant Zombie	Overworld, night	Slower but tankier. Ground-pound AoE attack. Summons regular zombies	Hulk Hammer (AoE knockback weapon)
Mutant Skeleton	Overworld, night	Rapid-fire arrows that inflict weakness. Teleports to escape melee	Mutant Skeleton Armor (projectile resistance)
Mutant Creeper	Overworld, night	Explosion radius 2× larger. Drops mini-creepers on death that also explode	Creeper Minion (craft a creeper that follows and explodes on command)
Mutant Enderman	The End	Teleports more aggressively. Throws blocks. Takes reduced damage from projectiles	Enderman gear (teleportation-related items)

Spawn rate config: If mutants are too frequent (or too rare), adjust in Mod Menu → Mutant Monsters → spawn weight.

Strategy: Mutants are vulnerable to the same tactics as their vanilla versions. Mutant Zombie → kite. Mutant Skeleton → close distance fast. Mutant Creeper → ranged attacks. Mutant Enderman → water buckets and fire resistance.

L_Ender's Cataclysm — Epic Boss Fights

Epic, hand-animated boss fights with unique mechanics and powerful loot. Each boss is **opt-in** — you must build a summoning structure or find a specific location.

Boss	Location	How to Summon	HP	Recommended Gear
Leviathan	Deep ocean (Y < 40)	Build a summoning altar from prismarine + sea lanterns in a deep ocean biome. Right-click with a Prismarine Shard	~200 HP	Diamond tier, Depth Strider boots, door trick (block line-of-sight projectiles)
Netherite Monstrosity	Nether wastes	Find the lair structure (large nether brick building). Enter the central arena	~300 HP	Full diamond/netherite, fire resistance potion
Ender Guardian	End (outer islands)	Defeat the Ender Dragon first. Find the summoning platform on an outer island. Place an Eye of Ender in the pedestal	~400 HP	End-game gear. Elytra + Slow Falling for arena edge
Ancient Remnant	Desert	Find the Remnant structure (large desert pyramid variant). Enter the summoning room. Insert a Nether Star	~350 HP	Blast protection enchantments. Bring blocks to pillar
Ignite	Nether (basalt deltas)	Build a hollow obsidian sphere (5×5 interior). Place a Blaze Rod in the center. Wait 10 seconds	~250 HP	Fire resistance required. Bring water buckets for arena

General boss strategy tips:

1. **Scout the arena first** — know where the edges, pits, and hazards are
2. **Bring food buffs** — Farmer's Delight meals + Brewin' And Chewin' drinks stack
3. **Rune up** — apply Lifesteal or Ice runes for the fight
4. **Set a Waystone nearby** — if you die, your grave holds items (GraveStone Mod). Re-equip and re-engage
5. **AE2 wireless terminal** — call in materials mid-fight if you need blocks, potions, or replacement gear

Leviathan — Deep Ocean Boss

- **Phase 1:** Charges at you. Dodge sideways, attack the body. It thrashes — jump over the shockwave
- **Phase 2** (50% HP): Summons whirlpools. Stay out of them — they pull you under. Attack the head between whirlpool spawns
- **Drops:** Leviathan Blade (trident-like weapon with water affinity), Sea-themed gear materials

Netherite Monstrosity — Nether Boss

- **Phase 1:** Slow melee swings. Attack the legs to slow it down
- **Phase 2** (60% HP): Enrages — faster attacks, fire breath. Minecart rails on the floor? Not here. Sprint behind pillars to block line of sight
- **Drops:** Monstrosity tools (efficient mining on netherite-level), Netherite gear upgrades

Ender Guardian — End Boss

- **Phase 1:** Ranged energy bolts. Dodge them — they track slightly. Hit the crystals on the arena walls to stun it
- **Phase 2** (50% HP): Teleports around. Summons endermates. Break all crystals to lower its shield
- **Phase 3** (20% HP): Enraged — faster teleports, energy nova. Pillar up to avoid the ground AOE's
- **Drops:** Ender Guardian weapons (high damage), Teleportation gear

Ancient Remnant — Desert Boss

- **Phase 1:** Sandstorm attack — blinds you. Wear goggles or turn away. Attack during cooldown
- **Phase 2** (40% HP): Sinks into sand, pops up elsewhere. Shoot the exposed weak point
- **Drops:** Ancient tools with unique enchantability, Desert-themed gear

Ignite — Fire Boss

- **Phase 1:** Fireball barrage. Block with a shield or dodge behind pillars. Attack after each barrage
- **Phase 2** (40% HP): Absorbs lava from the arena — stay on high ground. Attack when it emerges
- **Drops:** Fire weapons (permanent fire aspect without enchantment), Blaze-themed gear

When Dungeons Arise — Roguelike Structures

Large, multi-floor combat structures scattered across the Overworld. Each is a hand-built dungeon with increasing difficulty per floor.

Structure Types

Structure	Floors	Difficulty	Biome	Loot
Abandoned Temple	3	Easy	Plains, forests	Iron gear, early rare items
Small Castle	4	Medium	Plains, hills	Rare weapons, gold
Large Castle	6	Hard	Mountains, extreme hills	Epic gear, diamonds
Barracks	2	Easy	Any	Armor, shields
Scorched Fort	4	Medium	Desert, badlands	Fire-related items
Warped Fortress	5	Hard	Any (rare)	End-game gear, unique loot
Foundry	3	Medium	Any underground	Ores, mining gear
Bandit Camp	1	Easy	Forest, taiga	Early weapons, food

Finding structures: Craft an Explorer's Compass → select "When Dungeons Arise" in the structure type filter → select the specific structure.

Strategy: Each floor has a mob spawner or boss spawn point. Clear floors systematically. Loot scales with floor depth — deeper floors in large castles have the best loot. Bring blocks to pillar up, torches to light hallways, and decent gear for the deeper floors.

Tip: Structures are finite per world — once looted, they don't respawn unless you're far enough away for chunks to reset. Use the compass to find unvisited structures. Mark looted ones on your map.

Create Big Cannons — Artillery Engineering

Create Big Cannons adds functional cannons built from Create materials, using rotational power for assembly and gunpowder for ammunition.

Getting Started

1. **Craft cannon components:** Cannon barrel (bronze/steel), cannon mount (Create materials), cannon charger
2. **Build the cannon:** Place the mount, attach the barrel, connect the charger
3. **Arm it:** Insert powder charges and cannonballs into the cannon's GUI
4. **Fire:** Right-click the cannon or send a redstone signal

Ammunition Types

Ammo	Effect	Use case
Solid cannonball	High damage, block-breaking explosive	Siege, defense
Canister shot	Spread of small projectiles	Anti-personnel, mob clearing
Grape shot	Medium spread	Crowd control
Incendiary	Fire on impact + lingering fire	Area denial
Smoke shell	Smoke screen	Signalling, concealment

Integration

- Cannons are gated behind bronze/steel (Create's progression) and gunpowder (early game)
- Mount cannons on your colony's walls for defense — they fire at enemies within range
- Use cannonballs for large-scale terraforming (clearing areas for expansion)
- Redstone-controlled: hook up a pulse extender for automatic salvo fire

Session Plan — Wave 5

Combat content is opt-in. Engage when you want a challenge, not before.

Session	Focus	What to Do
Session 1 — Combat feel	Better Combat works automatically. Craft a Simply Swords weapon if you haven't already (or find one from dungeon loot). Test the attack patterns — daggers stab fast, greatswords sweep wide. Equip a Lifesteal or Fire rune	Learn your weapon, feel the combat change
Session 2 — First mutant	Find a Mutant Zombie at night (listen for its distinctive roar). Kite it, learn its attack patterns. Collect its unique drop	First mini-boss beaten, unique loot
Session 3 — First dungeon	Find a When Dungeons Arise small castle or abandoned temple. Clear floors systematically. Collect loot from each floor	Rougelike dungeon completed, rare loot

Session	Focus	What to Do
Session 4 — First Cataclysm boss	Choose the easiest boss (Leviathan — deep ocean) or the most accessible (Netherite Monstrosity — Nether fortress). Prepare: bring good weapons, fire resistance, healing food. Summon and fight	Epic boss beaten, endgame gear start
Session 5 — Cannons	Craft bronze (Create: zinc + copper in heated mixer). Build a Small Cannon. Test fire it at a target. Mount it near your base entrance	Artillery online, colony defense
Session 6+ — Boss gauntlet	Work through remaining Cataclysm bosses. Each drops unique materials for endgame gear. Twilight Forest bosses (Wave 2) also benefit from Wave 5 gear — revisit them now	Full endgame gear progression

Tip: Better Combat changes how melee works fundamentally. Don't skip Session 1 even if you're a ranged player — the weapon swing changes affect all melee combat, including tools.

Wave 6 — Building & Decoration

Tools and blocks to build the base you deserve. Every base deserves to look good — these mods give you the materials and tools to build without creative mode.

Install

See [MODLIST-V3.md](#).

Chipped — Infinite Block Variants

Chipped adds hundreds of decorative block variants for every material type. Every common block (stone, wood, glass, wool, terracotta, concrete, etc.) gets 8-50+ variants each.

How to Use — The Sawmill

1. **Craft a sawmill:** Any wood planks + iron ingot (check JEI for exact shape). It's a functional block, not decorative
2. **Place the sawmill** on the ground
3. **Right-click to open the GUI**
4. Place a base block in the **left slot** (e.g., stone)
5. The sawmill shows all available variants for that block type in the output panel
6. Click one to craft it. It consumes the input block and produces the variant

Block types with many variants:

- **Stone:** Bricks, tiles, carved, pillars, panels, runed, chiseled, mosaic — 30+ total
- **Glass:** All colors + clear, leaded, bordered, checkered, raised — perfect for windows
- **Wood:** Bark, planks with grain variants, wooden panels, ornate carvings
- **Wool & Carpet:** Woven, knitted, cross-stitched, patterned — for cozy floors
- **Concrete:** Smooth, paneled, tiles, with borders

- **Nether materials:** Glowstone, nether bricks, blackstone get many variants

Pro tip: The sawmill preserves block IDs for AE2 compatibility. You can store Chipped variants in your ME drive like any other block. Use AE2's pattern provider to autocraft specific variants on demand.

Supplementaries — Vanilla+ Functional Decoration

Supplementaries adds decoration blocks that serve a purpose:

Block	Function	How to use
Jar	Stores 1 item visually (display)	Right-click with an item to put it inside. Right-click empty to take it out. Shows the item floating inside
Signpost	Directional sign pointing to a named location	Right-click with a tag item to set the destination name. The signpost arm points toward that named Waystone or location
Faucet	Fluid flow decoration. Connects to fluid tanks	Place above a fluid container. Right-click to toggle flow. Decorative but functional with a fluid mod
Weather Vane	Shows wind direction and speed	Place outside. The vane rotates in real-time based on weather. Purely decorative
Item Shelf	Wall-mounted shelf for displaying up to 3 items	Right-click with an item to place it on the shelf. Sneak-right-click to remove. Items sit visibly on the shelf
Rope	Climbable rope, can be placed in mid-air	Right-click to place a rope knot. Add rope below by right-clicking with more rope. Climb like a ladder. Breaks when the top knot is broken
Flute	Craftable instrument	Right-click to play a note. Decorative, but villagers react to it

Practical builds:

- Kitchen: Item shelves for ingredient display, jars for flour/sugar, faucet over sink
- Entrance: Signposts directing to "Mine", "Nether Portal", "Village"
- Garden: Weather vane for realism, rope swing from a tree branch
- Study: Jars with glowing items as lamps

Macaw's — Complete Building Set

Macaw's adds structural building elements in every wood type. Five modules:

Macaw's Bridges:

- Craft: 6 slabs of the wood type + 1 stick in a bridge pattern
- Place: Right-click to extend. Shift-right-click to shorten. Supports railings, torches on the side
- Types: Straight bridge, rope bridge (no railings, more rustic)
- All wood types: oak, spruce, birch, jungle, acacia, dark oak, mangrove, cherry, crimson, warped

Macaw's Doors:

- Adds **special doors** per wood type: glass door, slatted door, stable door (top half opens independently), cottage door, shop door, paper door
- All crafted with matching wood + relevant materials (glass for glass doors, paper for paper doors)

Macaw's Roofs:

- Adds **roofing blocks** for every wood: roof slopes, ridge blocks, gutter blocks
- Craft: 3 slabs in a triangle pattern for a slope piece
- Place roofs by laying slope pieces side-by-side. The game auto-connects corners

Macaw's Windows:

- Window types: diamond pane, framed, barred, tropical, stained glass window
- All crafted with glass + wood type. Place like normal panes
- Can be combined with decorative glass from Chipped

Macaw's Fences:

- Variations beyond vanilla fences: picket fence, stockade fence, horse fence, wired fence, highley gate
- Each with a distinct look — picket for suburban, stockade for medieval, wired for industrial

Building tip: Use Macaw's roofs + Chipped stone variants + Macaw's windows for a cohesive base aesthetic. Create uses Macaw's trapdoors and fence gates for its kinetic builds, too.

Building Wands — Large-Scale Construction

Building Wands lets you place many blocks at once. It's the closest thing to WorldEdit without a server plugin.

Crafting: Stick + 2 blocks of the same type in a + pattern (check JEI — the blocks determine the wand tier; stone = basic, iron = better, diamond = best)

How to use:

1. **Right-click** the wand in your hand to open the **mode GUI**
2. Select a mode from the menu:

Mode	What it does	When to use
Build Wall	Creates a wall between 2 clicked points	Enclosing a room, building a perimeter
Build Line	Creates a 1-block-wide line between 2 points	Railings, borders, pathways
Fill Area	Fills a 3D cuboid between 2 corners	Filling in a foundation, clearing a room
Build Circle	Creates a ring of blocks	Towers, wells, circular rooms
Build Cylinder	Fills a cylinder	Silos, water tanks, pillar bases
Build Sphere	Creates a hollow or filled sphere	Domes, decorative globes

1. **Left-click** to set the first corner/point (a green sparkle appears)
2. **Right-click** to set the second corner/point (the structure is built)
3. **The wand uses blocks from your inventory** — it only places blocks you're carrying

Pro tips:

- Shift+Right-click to undo the last placement
- The wand works with Create schematic cannon blueprints for precision
- Use the **Undo wand** (craft: stick + slimeball) to reverse recent wand actions without losing materials
- Combined with AE2's ME terminal, you can access your entire storage while building — set a wireless terminal hotkey

Handcrafted — Furniture

Furniture blocks for building cozy interiors:

Furniture	What it does
Chairs	Right-click to sit. Looks different per wood type. Craft with slabs and planks
Tables	Decorative surface. Right-click to place items on top for display. Dinner tables, coffee tables, nightstands
Shelves	Wall-mounted. Right-click to place items. Books, potions, trophies
Curtains	Hanging window decoration. Right-click to open/close
Desks	Table + shelf combo. Right-click drawer for item storage
Cupboards	Small cabinet for aesthetic item storage. Must be placed against a wall
Sofa	Multi-block seating. Right-click to sit. Craft with wool + wood
Dresser	Tall storage furniture. Right-click drawers for storage

How to craft: Check JEI. Most furniture is crafted from wood + relevant material (wool for sofas, glass for cabinets). Different wood types give different styles.

Tip: Place Handcrafted furniture in the rooms you use most — kitchen, dining room, bedroom. Use Item Shelves for decorative displays of your favorite loot (Twilight Forest boss trophies, Apotheosis mythic gear, rare Simply Swords weapons).

Wave 7 — Space Exploration

Space travel becomes real. Build rockets, manage oxygen, explore alien worlds, and bring back exotic resources to fuel your industrial empire.

Install

See [MODLIST-V3.md](#).

Stellaris — Interplanetary Travel

Stellaris (by ExodusTeam) is a complete space exploration mod. Build rockets piece by piece, manage oxygen and fuel, and travel to 5 planets across 2 solar systems.

Core Concepts

Concept	What it means
Rocket Components	Rockets aren't crafted in one slot. Build engines, fuel tanks, capsules, and fins separately, then assemble them at a Rocket Station
Oxygen	Most planets have no atmosphere. You need a space suit with oxygen to survive. Oxygen runs out — refill at home before traveling
Fuel	Rockets need fuel (crafted from oil). Refine oil into fuel using Stellaris machines. Higher-tier rockets need more fuel
Planet Tiers	Planets have tier requirements — a basic rocket can only reach the Moon. Upgraded rockets can reach Mercury, Venus, Mars, and Glacio
Gravity	Each planet has different gravity. Lower gravity = higher jumps, slower falls. Higher gravity = weighted movement

Phase 1: Getting to Space

Step 1 — Gather early Stellaris resources:

- Find **oil** in the Overworld (generates as pools similar to water, often in deserts and plains)
- Mine **steel** — you already have this from Mekanism (Metallurgic Infuser)
- Collect **iron**, **copper**, **redstone**, and **glass** (all vanilla materials)

Step 2 — Build basic machines:

Machine	What it does	Priority
Rocket Station	The assembly block. Place it, open the GUI, build rocket components here	First
Fuel Refinery	Converts oil into rocket fuel. Place near the Rocket Station	Second
Oxygen Distributor	Fills space suits with oxygen. Requires power	Third
Water Separator	Produces oxygen from water (for the Oxygen Distributor)	Fourth
Solar Panel	Generates power on planets with sun exposure	Fifth

Step 3 — Build your first rocket:

1. Open the **Rocket Station** GUI
2. Craft a **Tier 1 Rocket Engine**, **Tier 1 Fuel Tank**, **Rocket Capsule**, and **Rocket Fins** using the station's internal crafting
3. Each component uses materials you've gathered: steel, iron, copper, redstone
4. Once all components are built, the Rocket Station assembles them into a **Tier 1 Rocket**
5. Right-click the assembled rocket with the **Rocket Station** to create the launch pad

Step 4 — Prepare for launch:

1. Fill the fuel tank — pump fuel from the Refinery into the rocket (fluid pipes from a pipe mod, or use buckets)
2. Equip a **Space Suit** (craft: check JEI — requires steel, glass, and rubber/leather)
3. Fill the space suit with oxygen at the **Oxygen Distributor**
4. Stock supplies — food (Farmer's Delight meals work with oxygen present), building blocks, torches, weapons

Step 5 — Launch:

1. Enter the rocket (right-click the door)
2. Press the launch button inside the rocket GUI
3. Watch the ascent animation — you're going to space

The Moon — Your First Destination

The Moon is the safest first destination. Gravity is low ($\approx 17\%$ of Overworld), no atmosphere (needs oxygen), and has basic resources.

What to do on the Moon:

- **Mine moon stone** and **cheese ore** (yes, the moon is made of cheese — it smelts into cheese ingots)
- **Collect desh** — the Moon's primary rare resource. Desh is used for Tier 2 rocket components
- **Explore structures** — crashed spaceships, moon bases, landing pads (Stellaris generates these automatically)
- **Set up a base** — place a Waystone immediately. A solar panel + oxygen distributor + chest is enough for early visits
- **Look for rich ore veins** — the Moon has higher concentrations of certain Overworld ores

Returning to Earth: Enter the rocket on the Moon and launch. You'll fly back to your original launch pad.

Other Planets

Planet	Tier Required	Gravity	Hazards	Key Resource	Notes
Moon	1	Low	None (oxygen required)	Desh	First destination. Safe. Practice here
Mercury	2	Low	Extreme heat, fire damage outside	Ostranium	Closest to the sun. Bring fire resistance
Venus	2	High	Thick atmosphere, acid rain	Tharsite	Dense, heavy movement. Bring acid protection
Mars	3	Medium	Cold, dust storms	Desh (rich veins)	Alien flora and fauna. Best for a permanent off-world base
Glacio	4	Low	Extreme cold, freezing damage	Glacio crystals	Ice planet. High-end resources. Endgame destination

Machines to Build Off-World

Once you have a foothold on another planet, set up automated resource extraction:

Machine	What it does	Why you need it
Solar Panel	Generates power from sunlight	Most planets have more sun than Earth. Free energy
Oxygen Distributor	Fills space suits with breathable air	Required for survival on every planet
Water Separator	Produces oxygen from water	Feed oxygen into the Distributor. Water pumps on some planets
Fuel Refinery	Processes oil into rocket fuel	Fuel is heavy — refine on the destination planet if you find oil
Pump Jack	Extracts oil from the ground	Find oil pools on other planets for local fuel production

Integration With Your Tech Base

System	How it connects
Mekanism power	Run Mekanism generators at your Moon base. Use universal cables to power Stellaris machines. The MekaSuit's flight and environmental protection make spacewalks trivial
Mekanism ore processing	Bring planet ores (desh, ostranium, tharsite) back to your main Mekanism processing line. 5x multiplication works on Stellaris ores
AE2 storage	Store Stellaris materials in your ME drive. Set up autocrafting for rocket components. Place an ME interface + storage bus at your off-world base with a quantum link or P2P tunnel to the main network
Create transport	Use Create trains to move rocket materials from the launch pad area to your main base. Conveyor belts move components from storage to the Rocket Station
TFMG fuel	Refine diesel/gasoline in TFMG's Distillation Tower → use it as rocket fuel for Stellaris (requires TFMG-Stellaris Compatibility). Single fuel chain from oil well to launch pad
Sophisticated Backpacks	Backpacks with netherite upgrades carry enough oxygen tanks and supplies for extended planetary expeditions
Farmer's Delight	Cooked meals are eatable in space (with oxygen present) — bring hearty stews and sandwiches instead of raw ingredients

Wave 7 Addons

Three additions extend Stellaris beyond its core experience:

Potentials API — Required Library

Potentials API is a mandatory dependency for Stellaris. Install it without reservation — it provides networking and capability code that Stellaris's rocket systems, oxygen management, and planetary mechanics rely on. No user-facing features, no config, no gameplay changes.

No config needed — library mod, install and forget.

TFMG-Stellaris Compatibility — Fuel Bridge

The **Create: TFMG - Stellaris Compatibility** mod bridges TFMG's oil refining with Stellaris's rocket fuel system. Without this mod, Stellaris has its own fuel crafting recipe (oil → fuel refinery). With it, you can use TFMG's diesel and gasoline as rocket fuel directly.

What it changes:

- TFMG diesel can fuel Stellaris rockets (no need for Stellaris's separate fuel refinery if you have TFMG's Distillation Tower)
- TFMG gasoline also works as rocket fuel (lower efficiency than diesel)
- Stellaris's own fuel recipe remains available as a fallback
- Single fuel chain: find oil → TFMG Distillation Tower → diesel → rocket fuel

Config: No changes needed. Compatibility works automatically once both TFMG and Stellaris are installed.

Progression Placement

Space exploration belongs in Phase 2 (Industrial Ascent, hours 40–200). You should have:

- Mechanism basic processing (Enrichment Chamber — Tier 1)
- A small AE2 network for storage
- Create power infrastructure
- Good weapons and armor (Wave 3/5 gear helps with planet mobs)

Start with moon missions, establish a Tier 1 base, then work toward Mercury and Venus. Mars and Glacio are solidly in late Phase 2 territory.

Session Plan — Wave 7

Space is a Phase 2+ activity. Don't start until you have Mechanism basic processing, an AE2 network, and decent power infrastructure.

Session	Focus	What to Build
Session 1 — Ground prep	Find oil (surface pools in deserts/plains). Build Fuel Refinery + Rocket Station. Craft Tier 1 rocket components (engine, fuel tank, capsule, fins). Assemble the rocket	Launch infrastructure complete
Session 2 — Moon landing	Craft a Space Suit. Fill it with oxygen at the Oxygen Distributor. Stock supplies (food, blocks, torches, weapons). Enter the rocket and launch	First off-world base, desh collected
Session 3 — Moon base	Set up a solar panel + oxygen distributor on the Moon. Place a Waystone. Mine desh for Tier 2 rocket components. Place a	Sustainable Moon operation

Session	Focus	What to Build
	Chunk Loader to keep it running	
Session 4 — Next planets	Upgrade to Tier 2 rocket. Fly to Mercury or Venus. Each has unique hazards (heat, acid rain) and resources (ostranium, tharsite). Set up a temporary base	New planet resources, expanded capabilities
Session 5+ — Deep space	Tier 3 rocket → Mars (permanent off-world base). Tier 4 rocket → Glacio (endgame crystals). Bring Mekanism digital miner for automated resource extraction	Full interplanetary logistics

Tip: Space is the most resource-intensive activity in the pack. Don't rush it. Each planet should feel like a meaningful expansion of your industrial empire, not a checklist item. The MekaSuit trivializes spacewalks — jetpack flight + environmental protection = no oxygen worries.

Keybindings Reference

Keybind	Default	Mod	Wave
View Recipes	R	JEI	0
View Uses	U	JEI	0
Open Mod Menu	Set to Ctrl+M	Mod Menu	0
Open Fullscreen Map	J	Xaero's World Map	0
Toggle Minimap	H	Xaero's Minimap	0
Open Waystone Menu	Set to Ctrl+Shift+W	Waystones	0
Open Backpack	B	Sophisticated Backpacks	0
Zoom	Set to C	Sodium	0
Toggle Dynamic Lights	Set to Ctrl+L	LambDynamicLights	0
Open Curios Inventory	Set to Y	Curios API	0
Open Skill Tree	Set to P	Skill Tree	3
View Weapon Ability	Ctrl (hold)	Simply Swords	3
Open Rune Pouch	Set to R	Runes	3
Open Create Ponder	W (hold over block)	Create	1
Open Town Hall GUI	Right-click Town Hall	MineColonies	1.5

Progression Through All Waves

Phase	What you'll be doing
Phase 1 (hours 0-40)	Wave 0 setup. Early Create (water wheel, mechanical press). Find certus quartz for AE2. Make first Mekanism machines (2x ore). Explore Terralith biomes. Skill Tree active from first action. Find your first Simply Swords weapon. Cook Farmer's Delight meals. Start MineColonies colony (builder hut, basic mines). Grab bounties from village boards
Phase 2 (hours 40-150)	Full Create factory (conveyors, deployers, sequenced crafters). Mekanism 3x-5x ore processing. AE2 ME network with autocrafting. Explore Twilight Forest bosses. Apotheosis rare/epic gear. Mutant Monsters. First Cataclysm boss. Build with Chipped and Macaw's. Complete bounties for passive rewards as you explore
Phase 3 (hours 150+)	Mekanism fusion reactor + MekaSuit. AE2 full autocrafting CPU chains. Endgame Apotheosis mythic gear. Twilight Forest final bosses. L_Ender's Cataclysm endgame bosses. BetterEnd End dimension. Monumental building projects with all decoration tools

General Tips

Finding Help

- **JEI (R)** is always your first stop — hover an item and press R to see how it's made
- **Mod Menu / Configured** shows every installed mod's config
- Most mods link to their wiki or Discord from their Mod Menu entry

Losing Progress

You won't. Graves preserve your items. No creeper craters. The pack is designed so you can walk away for a month and come back to exactly the world you left.

Java & Performance

Minecraft 1.21.1 requires **Java 21**. See [MODLIST-V3.md](#) for JDK installation instructions and recommended JVM arguments.